01 Indefinite Integration

If f & F are function of x such that

$$F'(x) = f(x)$$

then the function F is called a **PRIMITIVE OR ANTIDERIVATIVE OR INTEGRAL** of $f(x)$ w.r.t. x and is written symbolically as

$$\int f(x)dx = F(x)+C \Leftrightarrow \frac{d}{dx}\{F(x)+C\} = f(x),$$

where C is called the **constant of integration.**

1. GEOMETRICAL INTERPRETATION OF INDEFINITE INTEGRAL:

$\int f(x)dx = F(x)+C = y(say)$,

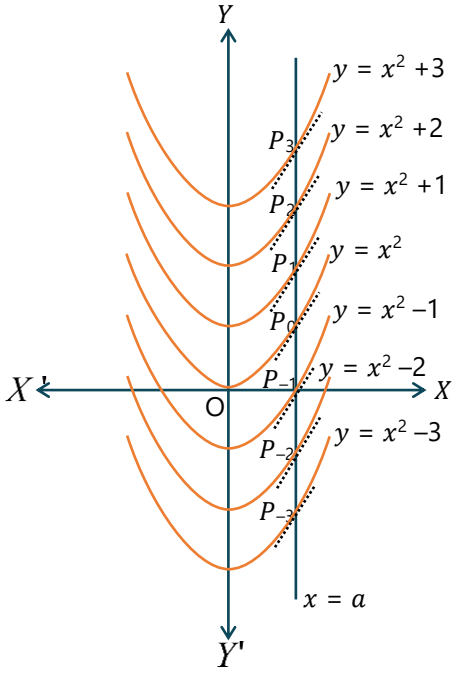
represents a family of curves. The different values of c will correspond to different members of this family and these members can be obtained by shifting any one of the curves parallel to itself. This is the geometrical interpretation of indefinite integral.

Let

$$f(x) = 2x. \text{ Then } \int f(x)dx = x^2 + C.$$

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For different values of C, we get different integrals. But these integrals are very similar geometrically. Thus, $y = x^2 + C$, where C is arbitrary constant, represents a family of integrals. By assigning different values to C, we get different members of the family. These together constitute the indefinite integral. In this case, each integral represents a parabola with its axis along y-axis.



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If the line $x = a$ intersects the parabolas $y = x^2$, $y = x^2 + 1$, $y = x^2 + 2$, $y = x^2 - 1$, $y = x^2 - 2$ at P_0, P_1, P_2, P_{-1}, P_{-2} etc., respectively, then $\frac{dy}{dx}$ at these points equals $2a$. This indicates that the tangents to the curves at these points are parallel. Thus, $\int 2x\,dx = x^2 + C = f(x) + C$ **(say), implies that the tangents to all the curves $f(x) + C$, $C \in R$, at the points of intersection of the curves by the line $x = a$, $(a \in R)$, are parallel.**

2. STANDARD FORMULAE:

We know that $\frac{d}{dx}\left(a^x\right) = a^x.\ell na \Rightarrow \int a^x dx = \frac{a^x}{\ell na} + C$ similarly,

$\frac{d}{dx}(\sin x) = \cos x \Rightarrow \int \cos x dx = \sin x + C$ and $\frac{d}{dx}(\cos x) = -\sin x \Rightarrow \int \sin x dx = -\cos x + C$

Proceeding in same way as above, we get the following formula.

(i) $\int x^n dx = \frac{x^{n+1}}{n+1} + C$; $n \neq -1$

(ii) $\int \frac{1}{x} dx = \ell n|x| + C$

(iii) $\int \cos x dx = \sin x + C$

(iv) $\int \sin x dx = -\cos x + C$

(v) $\int \sec^2 x dx = \tan x + C$

(vi) $\int \operatorname{cosec}^2 x\, dx = -\cot x + C$

(vii) $\int \sec x \tan x\, dx = \sec x + C$

(viii) $\int \operatorname{cosec} x \cot x\, dx = -\operatorname{cosec} x + C$

(ix) $\int a^x dx = \frac{a^x}{\ell na} + C(a > 0)$

(x) $\int e^x dx = e^x + C$

(xi) $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

(xii) $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$

(xiii) $\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1} x + C$

Note :

* If $f_1(x)$ and $f_2(x)$ are integrals of same function, then $f_1(x) - f_2(x)$ must be a constant.

* Constants can be taken out of integration i.e.

$$\int (kf(x))dx = k\int f(x)dx$$

* Integral of sum (difference) is equal to sum (difference) of integrals i.e.

$$\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$$

$$\int (f(x) - g(x))dx = \int f(x)dx - \int g(x)dx$$

Illustration 1

Find $\int \frac{x^2 - 2x + 3}{\sqrt{x}} dx$

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Solution:

$$\int \frac{x^2-2x+3}{\sqrt{x}}dx = \int \left(x^{3/2}-2x^{1/2}+3x^{-1/2}\right)dx$$

$$= \frac{x^{5/2}}{5/2}-\frac{2x^{3/2}}{3/2}+\frac{3x^{1/2}}{1/2}+C = \frac{2}{5}x^{5/2}-\frac{4}{3}x^{3/2}+6x^{1/2}+C$$

Illustration 2

Find $\int \frac{2^x+3^{2x+1}+5^{3x-2}}{7^x}dx$

Solution:

$$\int \frac{2^x+3^{2x+1}+5^{3x-2}}{7^x}dx$$

$$= \int \left(\left(\frac{2}{7}\right)^x+3\left(\frac{9}{7}\right)^x+\frac{1}{25}\left(\frac{125}{7}\right)^x\right)dx = \frac{\left(\frac{2}{7}\right)^x}{\ell n\left(\frac{2}{7}\right)}+3\frac{\left(\frac{9}{7}\right)^x}{\ell n\left(\frac{9}{7}\right)}+\frac{1}{25}\frac{\left(\frac{125}{7}\right)^x}{\ell n\left(\frac{125}{7}\right)}+C$$

Illustration 3

Find $\int \frac{x(\sin x+\cos x)+1}{x}dx$

Solution:

$$\int \frac{x(\sin x+\cos x)+1}{x}dx = \int \left(\sin x+\cos x+\frac{1}{x}\right)dx$$

$$= -\cos x+\sin x+\ell n|x|+C$$

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Illustration 4

Find $\int \frac{1+\sin^3 x+\cos^3 x}{\sin^2 x\cos^2 x}dx$

Solution:

$$\int \frac{1+\sin^3 x+\cos^3 x}{\sin^2 x\cos^2 x}dx$$

$$= \int \frac{\sin^2 x+\cos^2 x+\sin^3 x+\cos^3 x}{\sin^2 x\cos^2 x}dx$$

$$= \int \left(\sec^2 x+\text{cosec}^2 x+\sec x\tan x+\text{cosec}\,x\cot x\right)dx$$

$= \tan x - \cot x + \sec x - \text{cosec}\,x + C$

Illustration 5

Find $\int \frac{2\left(1+x^2\right)+3\sqrt{1-x^2}}{\left(1+x^2\right)\sqrt{1-x^2}}dx$

Solution:

$$\int \frac{2\left(1+x^2\right)+3\sqrt{1-x^2}}{\left(1+x^2\right)\sqrt{1-x^2}}dx = \int \left(\frac{2}{\sqrt{1-x^2}}+\frac{3}{1+x^2}\right)dx$$

$= 2\sin^{-1}x + 3\tan^{-1}x + C$

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3. EXTENSION OF STANDARD FORMULAE:

Recall the formulae we learnt in the previous section. In any of them, if x is replaced by $(ax + b)$; $a \neq 0$, then also the standard formula remains true, provided the result in divided by coefficient of x i.e. 'a', Here 'a' and 'b' are constants.

If $n \neq -1$, then $\int x^n dx = \frac{x^{n+1}}{n+1} + C \Rightarrow \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{(n+1)a} + C$

More generally

$$\int f(x)dx = F(x) + C \Rightarrow \int f(ax+b)dx = \frac{1}{a}F(ax+b) + C$$

For Example

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C;\ n \neq -1, \quad \int \frac{dx}{ax+b} = \frac{1}{a}\ell n|\ ax+b\ | + C$$

$$\int a^{px+q}dx = \frac{1}{p}\frac{a^{px+q}}{\ell na} + C,\ (a>0), \qquad \int \sin(ax+b)dx = -\frac{1}{a}\cos(ax+b) + C$$

$$\int \text{cosec}^2(ax+b)dx = -\frac{1}{a}\cot(ax+b) + C \ , \int \sec\ (ax+b).\tan(ax+b)dx = \frac{1}{a}\sec(ax+b) + C$$

Similarly, extension of other formulae can be obtained.

Illustration 6

Evaluate $\int\left(\frac{1}{8x+9} + e^{8x+9}\right)dx$

Solution:

$$\int \frac{1}{x}dx = \ell n|x| + C \Rightarrow \int \frac{dx}{8x+9} = \frac{1}{8}\ell n|8x+9| + C$$

$$\int e^x dx = e^x + C \Rightarrow \int e^{8x+9}dx = \frac{e^{8x+9}}{8} + C$$

$$\therefore \int\left(\frac{1}{8x+9} + e^{8x+9}\right)dx = \frac{\ell n|8x+9| + e^{8x+9}}{8} + C$$

Illustration 7

Find $\int\left[\cos(5+4x) + \sec^2(3-4x)\right]dx$

Solution:

$$\int \cos x dx = \sin x + C \Rightarrow \int \cos(5+4x)dx = \frac{\sin(5+4x)}{4} + C$$

Similarly $\int \sec^2(3-4x)dx = \frac{\tan(3-4x)}{-4} + C$

$$\therefore \int\left[\cos(5+4x) + \sec^2(3-4x)\right]dx = \frac{\sin(5+4x)}{4} - \frac{\tan(3-4x)}{4} + C$$

Illustration 8

Find $\int \frac{\cos 8x - \cos 7x}{1 + 2\cos 5x}dx$

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Solution:

$$\frac{\cos 8x-\cos 7x}{1+2\cos 5x}=\frac{-2\sin\frac{15x}{2}\sin\frac{x}{2}}{3-4\sin^2\frac{5x}{2}}\times\frac{\sin\frac{5x}{2}}{\sin\frac{5x}{2}}=\frac{-2\sin\frac{15\pi}{2}\sin\frac{x}{2}\sin\frac{5x}{2}}{\sin\frac{15x}{2}}$$

$$=-2\sin\frac{x}{2}\sin\frac{5x}{2}=\cos 3x-\cos 2x$$

$$\therefore\ \int\frac{\cos 8x-\cos 7x}{1+2\cos 5x}dx=\int(\cos 3x-\cos 2x)dx=\frac{\sin 3x}{3}-\frac{\sin 2x}{2}+C$$

Illustration 9

Evaluate $\int\frac{2}{(x-1)(x-4)}dx$

Solution:

$$\int\frac{2}{(x-1)(x-4)}dx$$

$$=\frac{2}{3}\int\frac{3}{(x-1)(x-4)}dx=\frac{2}{3}\int\frac{(x-1)-(x-4)}{(x-1)(x-4)}dx=\frac{2}{3}\int\left(\frac{1}{x-4}-\frac{1}{x-1}\right)dx=\frac{2}{3}\ell n\left|\frac{x-4}{x-1}\right|+C$$

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4. INTEGRALS BASED ON DIRECT SUBSTITUTION:

To evaluate integrals of form

$$I=\int f\left(g(x)\right).g'(x)dx$$

We substitute

$$g(x)=t\Rightarrow g'(x)=\frac{dt}{dx}\Rightarrow g'(x)dx=dt$$

Hence, we get

$$I=\int f\left(g(x)\right).g'(x)dx=\int f(t)dt \qquad .$$

Note : Here, $y = g(x)$ is differentiable function.

Illustration 10

Find $\int\frac{\ell nx^2}{x}dx$

Solution:

$$\int\frac{\ell nx^2}{x}dx=2\int\frac{\ell n|x|}{x}dx=2\int t\,dt \qquad [\text{Put } \ell n|x| = t\Rightarrow \frac{dx}{x}=dt\,]$$

$= t^2 + c = (\ell n|x|)^2 + C$

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Illustration 11

Find $\int \frac{\sin^{-1}x}{\sqrt{1-x^2}}dx$

Solution:

Let $\sin^{-1}x = t \Rightarrow \frac{1}{\sqrt{1-x^2}}dx = dt$

$\therefore \int \frac{\sin^{-1}x}{\sqrt{1-x^2}}dx = \int t\,dt = \frac{t^2}{2}+C = \frac{1}{2}(\sin^{-1}x)^2 + C$

Illustration 12

Find $\int \frac{\sin(\tan^{-1}x)}{1+x^2}dx$

Solution:

Put, $\tan^{-1}x = t \Rightarrow \frac{1}{1+x^2}dx = dt$

$\therefore \int \frac{\sin(\tan^{-1}x)}{1+x^2}dx = \int \sin t\,dt = -\cos t + C = -\cos(\tan^{-1}x) + C$

Illustration 13

Find $\int (4x+6)\sqrt{2x^2+6x+5}\,dx$

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Solution:

Let $2x^2 + 6x + 5 = t \Rightarrow (4x + 6)dx = dt$

$\therefore \int (4x+6)\sqrt{2x^2+6x+5}\,dx = \int \sqrt{t}dt = \frac{t^{3/2}}{3/2}+C = \frac{2}{3}\left(2x^2+6x+5\right)^{3/2}+C$

Illustration 14

Find $\int \frac{9x^2+2}{3x^3+2x+10}dx$

Solution:

Put $3x^3 + 2x + 10 = t \Rightarrow (9x^2 + 2)dx = dt$

$\therefore \int \frac{9x^2+2}{3x^3+2x+10}dx = \int \frac{dt}{t} = \ell n|t|+C = \ell n(3x^2+2x+10)+C$

Note : $\ell n|3x^2 + 2x + 10|$ is not used above as $3x^2 + 2x + 10 > 0\ \forall\ x \in R$

5. CONVERTIBLE TO DIRECT SUBSTITUTION:

Here, we will learn about integrals which can be converted into integrals involving direct substitution by simple manipulations. This is explained through the following illustrations.

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Illustration 15

Find $\int \frac{2x^3 dx}{1+x^2}$

Solution:

Put $1 + x^2 = t \Rightarrow 2xdx = dt$

$\therefore \int \frac{2x^3 dx}{1+x^2} = \int \frac{2x \cdot x^2 dx}{1+x^2} = \int \frac{(t-1)}{t} dt = \int \left(1 - \frac{1}{t}\right) dt$

$= t - \ell nt + C = (1 + x^2) - \ell n(1 + x^2) + C$

Illustration 16

Find $\int \frac{x^3}{\sqrt{x^2+2}} dx$

Solution:

$I = \int \frac{x^3}{\sqrt{x^2+2}} dx = \int \frac{x^2 (xdx)}{\sqrt{x^2+2}}$

Put $x^2 + 2 = t^2 \Rightarrow 2xdx = 2tdt \Rightarrow xdx = tdt$

$\therefore I = \int \frac{(t^2-2).(tdt)}{t} = \int (t^2-2) dt$

$= \frac{t^3}{3} - 2t + C = \frac{(x^2+2)^{\frac{3}{2}}}{3} - 2\sqrt{x^2+2} + C$

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Illustration 17

Find $\int \frac{2x^2+3x^5}{1+x^6} dx$

Solution:

$\int \frac{2x^2+3x^5}{1+x^6} dx = \frac{2}{3}\int \frac{3x^2 dx}{1+x^6} + \frac{3}{2}\int \frac{6x^5 dx}{1+x^6}$

In first integral, put

$x^3 = t \Rightarrow 3x^2 dx = dt$

In second integrals, put

$1 + x^6 = u \Rightarrow 6x^5 dx = du$

$\therefore \int \frac{2x^2+3x^5}{1+x^6} dx = \frac{2}{3}\int \frac{dt}{1+t^2} + \frac{1}{2}\int \frac{du}{u}$

$= \frac{2}{3}\tan^{-1}(t) + \frac{1}{2}\ell n|u| + C$

$= \frac{2}{3}\tan^{-1}(x^3) + \frac{1}{2}\ell n|1+x^6| + C$

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Illustration 18

Evaluate $\int \frac{x\left(1-x^2\right)}{1+x^4}dx$

Solution:

$$\int \frac{x\left(1-x^2\right)}{1+x^4}dx = \int\left(\frac{x}{1+x^4}-\frac{x^3}{1+x^4}\right)dx = \frac{1}{2}\arctan x^2 - \frac{1}{4}\ell n\left(x^4+1\right)+C$$

[Put $x^2 = t$ in first part and $x^4 + 1 = t$ in second part]

Illustration 19

Evaluate $\int \frac{2x-\sqrt{\arcsin x}}{\sqrt{1-x^2}}dx$

Solution:

$$\text{Let } I = \int \frac{2x-\sqrt{\sin^{-1}x}}{\sqrt{1-x^2}}dx = \int \frac{2xdx}{\sqrt{1-x^2}} - \int \frac{\sqrt{\sin^{-1}x}}{\sqrt{1-x^2}}dx$$

$$= \int \frac{-dt}{\sqrt{t}} - \int \sqrt{u}du \left[\begin{matrix}\text{Put } t = 1-x^2 \\ u = \sin^{-1}x\end{matrix}\right]$$

$$= -2\sqrt{t} - \frac{2}{3}u^{3/2}+C = -2\sqrt{1-x^2} - \frac{2}{3}\left(\sin^{-1}x\right)^{3/2}+C$$

6. SIMPLE SUBSTITUTIONS INVOLVING TRIGONOMETRIC INTEGRALS:

Integral	Substitution
$\int f(\sin x)\cos x\,dx$	Put $\sin x = t$
$\int f(\cos x)\sin x dx$	Put $\cos x = t$
$\int f(\tan x)\sec^2 x dx$	Put $\tan x = t$
$\int f(\cot x)\text{cosec}^2 x\,dx$	Put $\cot x = t$

Standard Formulae :

1. $\int \tan x dx = \ell n|\sec x| + C$

2. $\int \sec x dx = \ell n|\sec x + \tan x| + C$ or $\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+C$

3. $\int \cot x dx = -\ell n|\text{cosec } x|+C$

4. $\int \text{cosec} x dx = -\ell n|\cot x + \text{cosec} x| + C$ or $\ell n\left|\tan\frac{x}{2}\right|+C$

Illustration 20

Evaluate $\int \frac{\cos^3 x}{\sin^2 x+\sin x}dx$

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Solution:

$I = \int \frac{(1-\sin^2 x)\cos x}{\sin x(1+\sin x)}dx = \int \frac{1-\sin x}{\sin x}\cos x\,dx$

Put $\sin x = t \Rightarrow \cos x\,dx = dt$

$\Rightarrow I = \int \frac{1-t}{t}dt = \ell n|t| - t + C$

$= \ell n|\sin x| - \sin x + C$

Illustration 21

Find $\int \frac{\sin(\tan x)}{\cos^2 x}dx$

Solution:

Let $I = \int \frac{\sin(\tan x)}{\cos^2 x}dx = \int \sin(\tan x).\sec^2 x dx$

Put $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$\therefore I = \int \sin t dt = -\cos t + C = C - \cos(\tan x)$

Illustration 22

Find $\int \cot x\,dx$

Solution:

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$I = \int \cot x dx = \int \frac{\cos x dx}{\sin x}$

Put $\sin x = t \Rightarrow \cos x dx = dt$

$\therefore I = \int \frac{dt}{t} = \ell n|t| + C = \ell n|\sin x| + C$

Illustration 23

Find $\int \sec x dx$

Solution:

M-1

$I = \int \sec x dx = \int \frac{dx}{\cos x} = \int \frac{\cos x dx}{\cos^2 x}$

$= \int \frac{\cos x}{1-\sin^2 x}dx$

Put $\sin x = t \Rightarrow \cos x dx = dt$

$\therefore I = \int \frac{dt}{1-t^2} = \frac{1}{2}\ell n\left|\frac{1+t}{1-t}\right| + C$

$= \frac{1}{2}\ell n\left|\frac{1+\sin x}{1-\sin x}\right| + C$

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$= \frac{1}{2}\ell n\left|\frac{(1+\sin x)^2}{\cos^2 x}\right| + C$

$= \ell n\left|\frac{1+\sin x}{\cos x}\right| + C = \ell n|\sec x + \tan x| + C$ or $\ell n\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right| + C$

M-2

$I = \int \sec x dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx$

Put $\sec x + \tan x = t \Rightarrow (\sec x.\tan x + \sec^2 x)dx = dt$

$\therefore\ I = \int \frac{dt}{t} = \ell n|t| + C$

$= \ell n|\sec x + \tan x| + C$

Illustration 24

$\int \frac{e^{2x}(1+2x)}{\sin(xe^{2x})} dx$

Solution:

Put, $xe^{2x} = t \Rightarrow (e^{2x} + 2xe^{2x})\,dx = dt \Rightarrow e^{2x}(1+2x)\,dx = dt$

$\therefore\ \int \frac{e^{2x}(1+2x)}{\sin(xe^{2x})} dx = \int \frac{dt}{\sin t} = \int \text{cosec}\, t\, dt = \ell n\left|\tan\frac{t}{2}\right| + C = \ell n\left|\tan\frac{xe^{2x}}{2}\right| + C$

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Illustration 25

$\int \frac{\cos 2x}{\cos x} dx$

Solution:

$\int \frac{\cos 2x}{\cos x} dx = \int \frac{2\cos^2 x - 1}{\cos x} dx$

$= \int (2\cos x - \sec x)\, dx$

$= 2\sin x - \ell n|\sec x + \tan x| + C$

Illustration 26

$\int \text{cosec}\, x \ell n|\cot x - \text{cosec}\, x| dx$

Solution:

Put $\ell n|\cot x - \text{cosec}\, x)| = t \Rightarrow \text{cosec}\, x\, dx = dt$

$\therefore\ \int \text{cosec}\, x \ell n|\cot x - \text{cosec}\, x| dx = \int t\, dt = \frac{t^2}{2} + C = C + \frac{1}{2}\left(\ell n|\text{cosec}\, x - \cot x|\right)^2$

Illustration 27

$\int \frac{dx}{\sin(x+5)\sin(x+8)}$

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Solution:

Multiplying & dividing by sin(8 – 5)

$$\therefore I=\frac{1}{\sin(8-5)}\int\frac{\sin(8-5)}{\sin(x+5)\sin(x+8)}dx$$

$$=\frac{1}{\sin 3}\int\frac{\sin((x+8)-(x+5))\,dx}{\sin(x+5)\sin(x+8)}$$

$$=\frac{1}{\sin 3}\int\frac{\sin(x+8)\cos(x+5)-\cos(x+8)\sin(x+5)}{\sin(x+5)\sin(x+8)}dx$$

$$=\frac{1}{\sin 3}\int\left(\cot(x+5)-\cot(x+8)\right)dx=\frac{1}{\sin 3}.\ell n\left|\frac{\sin(x+5)}{\sin(x+8)}\right|+C$$

7. TRIGONOMETRIC INTEGRALS OF FORM:

$$\int\sin^m x\cos^n x\,dx$$

Case I : When m and n are whole numbers

* If one of them is odd, then substitute for the terms of even power.
* If both are odd, substitute either of the term.
* If both are even, use trigonometric identities to convert integrand into cosines of multiple angles.

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Case-II : $m + n$ is a negative even integer .

* In this case the best substitution is $\tan x = t$.

Illustration 28

Find $\int\sin^{10} x\cos^3 x dx$

Solution:

Let $I=\int\sin^{10} x\cos^3 x dx=\int\sin^{10} x\left(1-\sin^2 x\right)\left(\cos x dx\right)$

Put $\sin x = t \Rightarrow \cos x\,dx = dt$

$$\therefore\ I=\int t^{10}\left(1-t^2\right)dt=\int\left(t^{10}-t^{12}\right)dt$$

$$=\frac{t^{11}}{11}-\frac{t^{13}}{13}+C$$

$$=\frac{\sin^{11} x}{11}-\frac{\sin^{13} x}{13}+C$$

Illustration 29

Evaluate $\int\sin^2 x\cos^4 x dx$

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Solution:

$$\sin^2 x\cos^4 x=\frac{1}{4}\sin^2 2x.\cos^2 x$$

$$=\frac{1}{16}(1-\cos 4x)(1+\cos 2x)$$

$$=\frac{1}{16}(1+\cos 2x-\cos 4x-\cos 2x\cot x)$$

$$=\frac{1}{32}[2+\cos 2x-2\cos 4x-\cos 6x]$$

$$\therefore\ \int\sin^2 x\cos^4 x dx=\int\frac{1}{32}[2+\cos 2x-2\cos 4x-\cos 6x]dx$$

$$=\frac{x}{16}+\frac{\sin 2x}{64}-\frac{\sin 4x}{64}-\frac{\sin 6x}{192}+C$$

Illustration 30

Evaluate $\int\sin^3 x\cos^5 x\,dx$

Solution:

Method -1 :

Put $\cos x = t$; $-\sin x\,dx = dt$.

so that $I=\int\sin^3 x\cos^5 x\,dx=-\int(1-t^2).t^5 dt$

$$=\int(t^7-t^5)dt=\frac{t^8}{8}-\frac{t^6}{6}=\frac{\cos^8 x}{8}-\frac{\cos^6 x}{6}+C$$

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Method -2 :

Put $\sin x = t$; $\cos x\,dx = dt$

so that $I=\int t^3(1-t^2)^2 dt=\int\left(t^3-2t^5+t^7\right)dt$

$$=\frac{\sin^4 x}{4}-\frac{2\sin^6 x}{6}+\frac{\sin^8 x}{8}+C$$

Method -3 :

$$\sin^3 x\cos^5 x=\frac{1}{8}\sin^3 2x.\cos^2 x$$

$$=\frac{1}{64}\left(4\sin^3 2x\right)\left(2\cos^2 x\right)$$

$$=\frac{1}{64}(3\sin 2x-\sin 6x)(1+\cos 2x)$$

$$=\frac{1}{128}[6\sin 2x+6\sin 2x\cos 2x-2\sin 6x-2\sin 6x\cos 2x]$$

$$=\frac{1}{128}[6\sin 2x+5\sin 4x-2\sin 6x-\sin 8x]$$

$$\int\sin^3 x.\cos^5 x dx=\frac{1}{128}\left[\frac{\cos 6x}{3}+\frac{\cos 8x}{8}-\frac{5\cos 4x}{4}-\cos 2x\right]+C$$

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Illustration 31

Evaluate $\int \frac{\sqrt{\sin x}}{\cos^{9/2} x} dx$

Solution:

Let $I = \int \frac{\sin^{1/2} x}{\cos^{9/2} x} dx = \int \frac{dx}{\sin^{-1/2} x \cos^{9/2} x}$

Here $m + n = \frac{1}{2} - \frac{9}{2} = -4$ (negative even integer).

Divide Numerator & Denominator by $\cos^4 x$.

$I = \int \sqrt{\tan x} \sec^4 x\, dx = \int \sqrt{\tan x}(1+\tan^2 x)\sec^2 x dx$

$= \int \sqrt{t}(1+t^2)dt$ (using $\tan x = t$)

$= \frac{2}{3}t^{3/2} + \frac{2}{7}t^{7/2} + C = \frac{2}{3}\tan^{3/2} x + \frac{2}{7}\tan^{7/2} x + C$

Illustration 32

Evaluate $\int \frac{\sqrt{\tan x}}{\sin 2x} dx$

Solution:

Let $I = \int \frac{\sqrt{\tan x}}{2\tan x}(1+\tan^2 x)dx = \int \frac{\sqrt{\tan x}}{2\tan x}\sec^2 x\, dx = \int \frac{\sec^2 x\, dx}{2\sqrt{\tan x}}$

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Put $\tan x = t \Rightarrow \sec^2 x = \frac{dt}{dx}$

$\therefore I = \frac{1}{2}\int (t)^{-1/2} dt = \frac{t^{1/2} \times 2}{2 \times 1} + C = \sqrt{\tan x} + C$

8. INTEGRALS OF FORM:

$\int \frac{dx}{ax^2+bx+c}; a \neq 0, a, b, c \in R$

We proceed as follows :

$I = \int \frac{dx}{ax^2+bx+c} = \int \frac{dx}{a\left[\left(x+\frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a^2}\right]}$

Putting $x + \frac{b}{2a} = t$, we get $I = \frac{1}{a}\int \frac{dt}{t^2 \pm k^2}$.

Hence, to find $\int \frac{dx}{ax^2+bx+c}$, we need to remember the following two formulae :

1. $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c = \frac{-1}{a}\cot^{-1}\frac{x}{a} + c$
2. $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ell n\left|\frac{x-a}{x+a}\right| + c$

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Illustration 33

Find $\int \frac{dx}{x^2+a^2}$

Solution:

Let $I = \int \frac{dx}{x^2+a^2}$

Put $x = at \Rightarrow dx = a\,dt$

$\therefore\ I = \int \frac{a\,dt}{a^2t^2+a^2} = \frac{1}{a}\int \frac{dt}{t^2+1}$

$= \frac{1}{a}\tan^{-1}(t) + C = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

Illustration 34

Find $\int \frac{dx}{x^2-a^2}$

Solution:

$I = \int \frac{dx}{x^2-a^2} = \int \frac{2a}{2a\left(x^2-a^2\right)}dx = \frac{1}{2a}\int \frac{(x+a)-(x-a)}{(x+a)(x-a)}dx$

$= \frac{1}{2a}\int\left(\frac{1}{x-a} - \frac{1}{x+a}\right)dx = \frac{1}{2a}\left(\ell n|x-a| - \ell n|x+a|\right) + C$

$\therefore\ \int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ell n\left|\frac{x-a}{x+a}\right| + C$

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Illustration 35

Find $\int \frac{dx}{9x^2+4}$

Solution:

Let $I = \int \frac{dx}{9x^2+4}$

M-1 :

Put $3x = t \Rightarrow dx = \frac{dt}{3}$

$\therefore\ I = \int \frac{dt/3}{t^2+4} = \frac{1}{3}\int \frac{dt}{t^2+4}$

Now, use formula $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

$\therefore\ I = \frac{1}{3}\times\frac{1}{2}\tan^{-1}\frac{t}{2} + C = \frac{1}{6}\tan^{-1}\frac{3x}{2} + C$

M-2 :

Put $3x = 2t \Rightarrow dx = \frac{2}{3}dt$

$\therefore\ I = \int \frac{2dt}{3\left[4t^2+4\right]} = \frac{1}{6}\int \frac{dt}{t^2+1} = \frac{1}{6}\tan^{-1}(t) + C = \frac{1}{6}\tan^{-1}\frac{3x}{2} + C$

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Illustration 36

Evaluate $\int \frac{dx}{9-16x^2}$

Solution:

Put $4x = 3t \Rightarrow 4dx = 3dt$

$$\therefore \int \frac{dx}{9-16x^2} = \int \frac{3dt}{4\left[9-9t^2\right]} = \frac{-1}{12}\int \frac{dt}{t^2-1} = \frac{1}{24}\ell n\left|\frac{1+t}{1-t}\right| + C = \frac{1}{24}\ell n\left|\frac{3+4x}{3-4x}\right| + C$$

Illustration 37

Evaluate $\int \frac{dx}{3x^2+6x+15}$

Solution:

$$\int \frac{dx}{3x^2+6x+15} = \frac{1}{3}\int \frac{dx}{x^2+2x+5}$$

$$= \frac{1}{3}\int \frac{dx}{(x+1)^2+4} = \frac{1}{3}\int \frac{dx}{(x+1)^2+(2)^2} = \frac{1}{6}\tan^{-1}\left(\frac{x+1}{2}\right) + C$$

Illustration 38

Evaluate $\int \frac{dx}{2x^2+x-1}$

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Solution:

M-1 :

Let $I = \int \frac{dx}{2x^2+x-1} = \frac{dx}{(x+1)(2x-1)} = \frac{1}{3}\int \frac{2(x+1)-(2x-1)}{(x+1)(2x-1)}dx$

$$= \frac{1}{3}\int\left(\frac{2}{2x-1} - \frac{1}{x+1}\right)dx = \frac{1}{3}\ell n\left|\frac{2x-1}{x+1}\right| + C$$

M-2 :

$$I = \int \frac{dx}{2x^2+x-1} = \frac{1}{2}\int \frac{dx}{x^2+\frac{x}{2}-\frac{1}{2}} = \frac{1}{2}\int \frac{dx}{x^2+\frac{x}{2}+\frac{1}{16}-\frac{1}{16}-\frac{1}{2}}$$

$$= \frac{1}{2}\int \frac{dx}{\left(x+\frac{1}{4}\right)^2-\left(\frac{3}{4}\right)^2} = \frac{1}{2}\int \frac{dt}{t^2-a^2}\left[\begin{matrix} t = x+\frac{1}{4} \\ a = \frac{3}{4}\end{matrix}\right]$$

$$= \frac{1}{4a}\ell n\left|\frac{t-a}{t+a}\right| + C = \frac{1}{3}\ell n\left|\frac{x-\frac{1}{2}}{x+1}\right| + C = \frac{1}{3}\ell n\left|\frac{2x-1}{x+1}\right| + C$$

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9. INTEGRALS OF FORM:

$$\int \frac{1}{\sqrt{ax^2+bx+c}}dx; a \neq 0; a,b,c \in R$$

We proceed as follows :

$$I = \int \frac{dx}{\sqrt{ax^2+bx+c}} = \int \frac{dx}{\sqrt{a\left[\left(x+\frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a^2}\right]}}$$

Putting $x+\frac{b}{2a}=t$, we get

$$I = \int \frac{dt}{\sqrt{at^2+\frac{4ac-b^2}{4a}}}$$

Hence, to find $\int \frac{dx}{\sqrt{ax^2+bx+c}}$, we need to remember the following three formulae :

1. $\int \frac{dx}{\sqrt{x^2+a^2}} = \ell n\left|x+\sqrt{x^2+a^2}\right| + c$

2. $\int \frac{dx}{\sqrt{x^2-a^2}} = \ell n\left|x+\sqrt{x^2-a^2}\right| + c$

3. $\int \frac{1}{\sqrt{a^2-x^2}}dx = \sin^{-1}\frac{x}{a} + c = -\cos^{-1}\frac{x}{a} + c$

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9.1 Some Special Forms With Suggested Substitutions

Integral	Substitution
$\int \frac{dx}{\sqrt{a^2+x^2}}$ or $\int \sqrt{a^2+x^2}dx$	$x = a\tan\theta$ or $x = a\cot\theta$
$\int \frac{dx}{\sqrt{a^2-x^2}}$ or $\int \sqrt{a^2-x^2}dx$	$x = a\sin\theta$ or $x = a\cos\theta$
$\int \frac{dx}{\sqrt{x^2-a^2}}$ or $\int \sqrt{x^2-a^2}dx$	$x = a\sec\theta$ or $x = a\operatorname{cosec}\theta$
$\int \sqrt{\frac{a-x}{a+x}}dx$	$x = a\cos 2\theta$
$\int \sqrt{\frac{x-\alpha}{\beta-x}}dx$ or $\int \sqrt{(x-\alpha)(\beta-x)}dx$	$x = \alpha\cos^2\theta + \beta\sin^2\theta$
$\int \sqrt{\frac{x-\alpha}{x-\beta}}dx$ or $\int \sqrt{(x-\alpha)(x-\beta)}dx$	$x = \alpha\sec^2\theta - \beta\tan^2\theta$
$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$	$x-\alpha = t^2$ or $x-\beta = t^2$.

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Illustration 39

Find $\int \frac{dx}{\sqrt{x^2+a^2}}$

Solution:

Let $I = \int \frac{dx}{\sqrt{x^2+a^2}}$

Put $x = a\tan\theta \Rightarrow dx = a\sec^2\theta\, d\theta$

$\therefore\ I = \int \frac{a\sec^2\theta d\theta}{a\sec\theta} = \int \sec\theta\, d\theta$

$= \ell n\left|\sec\theta + \tan\theta\right| + C = \ell n\left|a\sec\theta + a\tan\theta\right| + C - \ell na$

$= \ell n\left|x + \sqrt{x^2+a^2}\right| + C \qquad [C - \ell na \text{ is a constant}]$

M-2

Put $x + \sqrt{x^2+a^2} = t \Rightarrow \left(1 + \frac{x}{\sqrt{x^2+a^2}}\right)dx = dt \Rightarrow \frac{dx}{\sqrt{x^2+a^2}} = \frac{dt}{t}$

$\therefore\ I = \int \frac{dx}{\sqrt{x^2+a^2}} = \int \frac{dt}{t} = \ell n\left|t\right| + C = \ell n\left|x + \sqrt{x^2+a^2}\right| + C$

Illustration 40

Find $\int \frac{dx}{\sqrt{a^2-x^2}}$

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Solution:

Let $I = \int \frac{dx}{\sqrt{a^2-x^2}}$

Put $x = a\sin\theta \Rightarrow dx = a\cos\theta\, d\theta$

$\therefore\ I = \int \frac{a\cos\theta d\theta}{a\cos\theta} = \int d\theta = \theta + C = \sin^{-1}\left(\frac{x}{a}\right) + C$

Illustration 41

Evaluate $\int \frac{dx}{\sqrt{4-x^2}}$

Solution:

Put $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\, d\theta$

$\therefore\ \int \frac{dx}{\sqrt{4-x^2}} = \int \frac{2\cos\theta d\theta}{2\cos\theta} = \theta + C = \sin^{-1}\frac{x}{2} + C$

Illustration 42

Evaluate $\int \frac{dx}{\sqrt{9-16x^2}}$

Solution:

Put $4x = t \Rightarrow \int \frac{dx}{\sqrt{9-16x^2}} = \int \frac{dt}{4\sqrt{9-t^2}} = \frac{1}{4}\sin^{-1}\frac{t}{3} + C = \frac{1}{4}\sin^{-1}\frac{4}{3}x + C$

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Illustration 43

Evaluate $\int \frac{dx}{\sqrt{3x^2+4}}$

Solution:

Put $\sqrt{3}x = t \Rightarrow \int \frac{dx}{\sqrt{3x^2+4}} = \int \frac{dt}{\sqrt{3}\sqrt{t^2+4}}$

$\Rightarrow \int \frac{dx}{\sqrt{3x^2+4}} = \frac{1}{\sqrt{3}} \ell n\left|\sqrt{3}x + \sqrt{3x^2+4}\right| + C$

Illustration 44

Find $\int \frac{dx}{\sqrt{x^2+4x+13}}$

Solution:

Let $I = \int \frac{dx}{\sqrt{x^2+4x+13}} = \int \frac{dx}{\sqrt{(x+2)^2+9}}$

Put $x + 2 = t \Rightarrow dx = dt$

$\therefore\ I = \int \frac{dx}{\sqrt{t^2+9}} = \ell n\left|t + \sqrt{t^2+9}\right| + C$

$= \ell n\left|x+2+\sqrt{x^2+4x+13}\right| + C$

Illustration 45

Find $\int \frac{dx}{\sqrt{12x-9x^2}}$

Solution:

Let $I = \int \frac{dx}{\sqrt{12x-9x^2}} = \int \frac{dx}{\sqrt{4-(3x-2)^2}}$

Put $3x - 2 = t \Rightarrow dx = \frac{dt}{3}$

$\therefore\ I = \int \frac{dt}{3\sqrt{4-t^2}} = \frac{1}{3}\sin^{-1}\left(\frac{t}{2}\right) + C = \frac{1}{3}\sin^{-1}\left(\frac{3x-2}{2}\right) + C$

Illustration 46

$\int \frac{2x}{\sqrt{x^4+2x^2+4}} dx$

Solution:

Put $x^2 = t \ \Rightarrow\ 2x\,dx = t$

$\therefore\ \int \frac{2x}{\sqrt{x^4+2x^2+4}} dx = \int \frac{dt}{\sqrt{t^2+2t+4}} = \int \frac{dt}{\sqrt{(t+1)^2+\left(\sqrt{3}\right)^2}}$

$= \ell n\left|(t+1) + \sqrt{(t+1)^2+\left(\sqrt{3}\right)^2}\right| + C = \ell n\left(\left(x^2+1\right) + \sqrt{x^4+2x^2+4}\right) + C$

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Illustration 47

Evaluate $\int \frac{dx}{\sqrt{(x-a)(b-x)}}$ $(b > a)$

Solution:

Put $x = a\cos^2\theta + b\sin^2\theta$. The given integral becomes

$$I = \int \frac{2(b-a)\sin\theta\cos\theta d\theta}{\left\{(a\cos^2\theta + b\sin^2\theta - a)(b - a\cos^2\theta - b\sin^2\theta\right\}^{\frac{1}{2}}}$$

$$= \int \frac{2(b-a)\sin\theta\cos\theta d\theta}{(b-a)\sin\theta\cos\theta} = \left(\frac{b-a}{b-a}\right)\int 2d\theta = 2\theta + C = 2\sin^{-1}\sqrt{\left(\frac{x-a}{b-a}\right)} + C$$

Illustration 48

Find $\int \sqrt{\frac{x-3}{5-x}}dx$

Solution:

Let $I = \int \sqrt{\frac{x-3}{5-x}}dx$

Put $x = 3\cos^2\theta + 5\sin^2\theta$

$\Rightarrow x - 3 = 2\sin^2\theta$, $5 - x = 2\cos^2\theta$

and $dx = 4\sin\theta\cos\theta\, d\theta$

$$\therefore\ I = \int \frac{\sin\theta}{\cos\theta} \times 4\sin\theta\cos\theta d\theta = 4\int \sin^2\theta d\theta$$

$$= 2\int (1 - \cos 2\theta) d\theta = 2\left[\theta - \frac{\sin 2\theta}{2}\right] + C$$

$$= 2\tan^{-1}\sqrt{\frac{x-3}{5-x}} - \sqrt{(x-3)(5-x)} + C$$

Illustration 49

Evaluate $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}.\frac{1}{x}dx$

Solution:

Put $x = t^2 \Rightarrow dx = 2tdt$

$$I = \int \sqrt{\frac{1-t}{1+t}} \times \frac{2tdt}{t^2} = \int \frac{2(1-t)dt}{t\sqrt{1-t^2}} = 2\int \frac{dt}{t\sqrt{1-t^2}} - 2\int \frac{dt}{\sqrt{1-t^2}} = 2\int \frac{dt}{t^2\sqrt{\frac{1}{t^2}-1}} - 2\sin^{-1}(t) + C$$

$$= 1\int \frac{-dx}{\sqrt{u^2-1}} - 2\sin^{-1}(t) + C \left[\begin{array}{l}\text{Put } \frac{1}{t} = u \\ \Rightarrow \frac{dt}{t^2} = -du\end{array}\right]$$

$$= -2\ell n\left|u + \sqrt{u^2-1}\right| - 2\sin^{-1}t + C = -2\ell n\left|\frac{1}{\sqrt{x}} + \sqrt{\frac{1}{x}-1}\right| - 2\sin^{-1}\sqrt{x} + C$$

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10. INTEGRALS OF FORM:

$$\int\left(\sqrt{ax^2+bx+c}\right)dx; a\neq 0, a,b,c\in R$$

We proceed as follows :

$$I=\int\sqrt{ax^2+bx+c}\,dx=\int\sqrt{a\left[\left(x+\frac{b}{2a}\right)^2+\frac{4ac-b^2}{4a^2}\right]}dx$$

Putting , $x+\frac{b}{2a}=t$ we get

$$I=\int\sqrt{at^2+\frac{4ac-b^2}{4a}}\,dt$$

Hence, to find $\int\sqrt{ax^2+bx+c}\,dx$, we need to remember the following three formulae :

1. $\int\sqrt{x^2+a^2}dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ell n\left|x+\sqrt{x^2+a^2}\right|+C$

2. $\int\sqrt{x^2-a^2}dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\ell n\left|x+\sqrt{x^2-a^2}\right|+C$

3. $\int\sqrt{a^2-x^2}dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$

Note : We will learn proof of above formulae after integration by parts, refer to illustration 88 and 89.

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Illustration 50

Evaluate $\int\sqrt{x^2+3}\,dx$

Solution:

Using the formula

$\int\sqrt{x^2+a^2}dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ell n\left|x+\sqrt{x^2+a^2}\right|+C$, we get

$$\int\sqrt{x^2+3}\,dx=\frac{x}{2}\sqrt{x^2+3}+\frac{3}{2}\ell n\left|x+\sqrt{x^2+3}\right|+C$$

Illustration 51

Evaluate $\int\sqrt{5-4x^2}dx$

Solution:

Using the formula

$\int\sqrt{a^2-x^2}dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$, we get

$$\int\sqrt{5-4x^2}dx=\frac{1}{4}\left(2x\sqrt{5-4x^2}+5\sin^{-1}\left(\frac{2x}{\sqrt{5}}\right)\right)+C$$

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Illustration 52

Find $\int\sqrt{x^2+2x+5}\,dx$

Solution:

Note that

$$\int\sqrt{x^2+2x+5}\,dx=\int\sqrt{(x+1)^2+4}\,dx$$

Put $x+1=y$, so that $dx=dy$. Then

$$\int\sqrt{x^2+2x+5}\,dx=\int\sqrt{y^2+2^2}\,dy\ =\frac{1}{2}y\sqrt{y^2+4}+\frac{4}{2}\log\left|y+\sqrt{y^2+4}\right|+C$$

$$=\frac{1}{2}(x+1)\sqrt{x^2+2x+5}+2\log\left|x+1+\sqrt{x^2+2x+5}\right|+C$$

Illustration 53

Find $\int\sqrt{3-2x-x^2}\,dx$

Solution:

Note that $\int\sqrt{3-2x-x^2}\,dx=\int\sqrt{4-(x+1)^2}\,dx$

Put $x+1=y$ so that $dx=dy$

Thus $\int\sqrt{3-2x-x^2}\,dx=\int\sqrt{4-y^2}\,dy=\frac{1}{2}y\sqrt{4-y^2}+\frac{4}{2}\sin^{-1}\frac{y}{2}+C$

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$$=\frac{1}{2}(x+1)\sqrt{3-2x-x^2}+2\sin^{-1}\left(\frac{x+1}{2}\right)+C$$

Illustration 54

Evaluate $\int\sqrt{e^{4x}+4e^{2x}}dx$

Solution:

Put $e^x=t\Rightarrow\int e^x\sqrt{e^{2x}+4}\,dx=\int\sqrt{t^2+4}\,dt=\frac{e^x}{2}\sqrt{e^{2x}+4}+2\ell n\left|e^x+\sqrt{e^{2x}+4}\right|+C$

Illustration 55

Evaluate $\int\sqrt{x^2-3x^6}dx\left(x>0\right)$

Solution:

$$\int\sqrt{x^2-3x^6}dx=\int\sqrt{1-3x^4}.xdx\ \text{[Put }\sqrt{3}x^2=t\ \Rightarrow\ 2\sqrt{3}xdx=dt\ ]$$

$$=\int\sqrt{1-t^2}\,\frac{dt}{2\sqrt{3}}\ =\frac{1}{4\sqrt{3}}\left[t\sqrt{1-t^2}+\sin^{-1}t\right]+C$$

$$=\frac{1}{4\sqrt{3}}\left[\sqrt{3}x^2\sqrt{1-3x^4}+\sin^{-1}\left(\sqrt{3}x^2\right)\right]+C$$

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11. INTEGRALS OF FORM:

$$\int \frac{px+q}{ax^2+bx+c}dx, \int \frac{px+q}{\sqrt{ax^2+bx+c}}dx, \int (px+q)\sqrt{ax^2+bx+c}\,dx; a \neq 0, a,b,c,p,q, \in R$$

Express $px + q = \ell$ (differential coefficient of denominator) + m.

Similarly, for integrals

$$\int \frac{ax^2+bx+c}{px^2+qx+r}dx, \int \frac{ax^2+bx+c}{\sqrt{px^2+qx+r}}dx$$

Express $ax^2 + bx + c = \ell(D^r) + m(D^r)' + n = \ell(px^2 + qx + r) + m(2px + q) + n$

Illustration 56

Evaluate $\int \frac{3x+2}{4x^2+4x+5}dx$

Solution:

Express $3x + 2 = \ell$(d.c. of $4x^2 + 4x + 5$) + m

or, $3x + 2 = \ell(8x + 4) + m$

Comparing the coefficients, we get

$8\ell = 3$ and $4\ell + m = 2 \Rightarrow \ell = 3/8$ and $m = 2 - 4\ell = 1/2$

$$\Rightarrow I = \int \frac{3x+2}{4x^2+4x+5}dx = \frac{3}{8}\int \frac{8x+4}{4x^2+4x+5}dx + \frac{1}{2}\int \frac{dx}{4x^2+4x+5}$$

$$= \frac{3}{8}\log\left|4x^2+4x+5\right| + \frac{1}{8}\int \frac{dx}{x^2+x+\frac{5}{4}}$$

$$= \frac{3}{8}\log\left|4x^2+4x+5\right| + \frac{1}{8}\tan^{-1}\left(x+\frac{1}{2}\right)+C$$

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Illustration 57

Evaluate $\int (2x-4)\sqrt{4+3x-x^2}dx$

Solution:

Write $(2x - 4) = \ell$ (d.c. of $4 + 3x - x^2$) + m or $(2x - 4) = \ell(-2x + 3) + m$.

Comparing the coefficient, we get

$-2\ell = 2, 3\ell + m = -4 \Rightarrow \ell = -1, m = -1$

Hence $I = \int (-2x+3)\sqrt{4+3x-x^2}dx - \int \sqrt{4+3x-x^2}dx$

$$= -\frac{2}{3}\left(4+3x-x^2\right)^{3/2} - \int \sqrt{\left(\frac{5}{2}\right)^2 - \left(x-\frac{3}{2}\right)^2}dx$$

$$= -\frac{2}{3}\left(4+3x-x^2\right)^{3/2} - \left[\frac{\left(x-\frac{3}{2}\right)}{2}\sqrt{\left(\frac{5}{2}\right)^2 - \left(x-\frac{3}{2}\right)^2} + \frac{25}{8}\sin^{-1}\left(\frac{2x-3}{3}\right)\right] + c$$

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Illustration 58

Evaluate $\int \frac{x^3+x^2+x+1}{\sqrt{x^2+2x+3}}dx$.

Solution:

$x^3 + x^2 + x + 1 = x(x^2 + 2x + 3) - 1(x^2 + 2x + 3) + 4$

Hence $\int \frac{x^3+x^2+x+1}{\sqrt{x^2+2x+3}}dx = \int (x-1)\sqrt{x^2+2x+3}\,dx + \int \frac{4}{\sqrt{x^2+2x+3}}dx$

$= \frac{1}{2}\int (2x+2)\sqrt{x^2+2x+3}\,dx - 2\int \sqrt{x^2+2x+3}\,dx + \int \frac{4}{\sqrt{x^2+2x+3}}dx$

$= \frac{1}{2}\times\frac{2}{3}\left(x^2+2x+3\right)^{3/2} - (x+1)\sqrt{x^2+2x+3} + 2\ell n\left|x+1+\sqrt{x^2+2x+3}\right| + C$

12. ALGEBRAIC TWINS:

Integrals of the form $\int \frac{x^2+1}{x^4+Kx^2+1}dx$ OR $\int \frac{x^2-1}{x^4+Kx^2+1}dx$, where K is any constant.

Divide N^r & D^r by x^2 & proceed.

Note : Sometimes it is useful to write the integral as a sum of two related integrals, which can be evaluated by making suitable substitutions e.g.

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* $\int \frac{2x^2}{x^4+1}dx = \int \frac{x^2+1}{x^4+1}dx + \int \frac{x^2-1}{x^4+1}dx$

* $\int \frac{2}{x^4+1}dx = \int \frac{x^2+1}{x^4+1}dx - \int \frac{x^2-1}{x^4+1}dx$

These integrals can be called as **Algebric Twins**.

The major concept here is the following manipulation

$x^2 + \frac{1}{x^2} = \left(x+\frac{1}{x}\right)^2 - 2$ and $\left(x-\frac{1}{x}\right)^2 + 2$

The ability to convert $x^2+\frac{1}{x^2}$ into a function of $x+\frac{1}{x}$ or $x-\frac{1}{x}$, provides an opportunity for substitution in case numerator contains derivative of any of them i.e. $\left(1-\frac{1}{x^2}\right)$ or $\left(1+\frac{1}{x^2}\right)$ respectively.

Illustration 59

Evaluate $\int \frac{(x^2-1)dx}{(x^4+3x^2+1)\tan^{-1}\left(x+\frac{1}{x}\right)}$

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Solution:

The given integral can be written as $I=\int\frac{\left(1-\frac{1}{x^2}\right)dx}{\left(x^2+\frac{1}{x^2}+3\right)\tan^{-1}\left(x+\frac{1}{x}\right)}$

$I=\int\frac{\left(1-\frac{1}{x^2}\right)dx}{\left[\left(x+\frac{1}{x}\right)^2+1\right]\tan^{-1}\left(x+\frac{1}{x}\right)}$

Let $\left(x+\frac{1}{x}\right)=t\Rightarrow\left(1-\frac{1}{x^2}\right)dx=dt$

Hence $I=\int\frac{dt}{(t^2+1)\tan^{-1}t}$

Let $\tan^{-1}t=u$. Then $\frac{dt}{t^2+1}=du$ and $I=\int\frac{du}{u}=\ell n|u|+C$

$I=\ell n|\tan^{-1}t|+C=\ell n\left|\tan^{-1}\left(x+\frac{1}{x}\right)\right|+C$

Illustration 60

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Find $\int\frac{1+x^2}{1+x^4}dx$

Solution:

Let $I=\int\frac{1+x^2}{1+x^4}dx=\int\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$

Put $x-\frac{1}{x}=t\Rightarrow x^2+\frac{1}{x^2}=t^2+2$ and $\left(1+\frac{1}{x^2}\right)dx=dt$

$\therefore\ I=\int\frac{dt}{t^2+2}=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)+C$

$=\frac{1}{2}\tan^{-1}\left(\frac{x^2-1}{x\sqrt{2}}\right)+C$

Illustration 61

Find $\int\frac{1-x^2}{1+x^4}dx$

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Solution:

Let $I=\int\frac{1-x^2}{1+x^4}dx=\int\frac{\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}}dx$

Put $x+\frac{1}{x}=t\Rightarrow x^2+\frac{1}{x^2}=t^2-2$ and $\left(1-\frac{1}{x^2}\right)dx=dt$

$\therefore\ I=\int\frac{-dt}{t^2-2}=\frac{1}{2\sqrt{2}}\ell n\left|\frac{t+\sqrt{2}}{t-\sqrt{2}}\right|+C$

$=\frac{1}{2\sqrt{2}}\ell n\left|\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right|+C$

Illustration 62

Evaluate : $\int\frac{1}{x^4+5x^2+1}dx$

Solution:

Let $I=\frac{1}{2}\int\frac{2}{x^4+5x^2+1}dx$

$\Rightarrow I=\frac{1}{2}\int\frac{1+x^2}{x^4+5x^2+1}dx+\frac{1}{2}\int\frac{1-x^2}{x^4+5x^2+1}dx=\frac{1}{2}\int\frac{1+\frac{1}{x^2}}{x^2+5+\frac{1}{x^2}}dx-\frac{1}{2}\int\frac{1-\frac{1}{x^2}}{x^2+5+\frac{1}{x^2}}dx$

{dividing N^r and D^r by x^2}

$=\frac{1}{2}\int\frac{\left(1+\frac{1}{x^2}\right)}{\left(x-\frac{1}{x}\right)^2+7}dx-\frac{1}{2}\int\frac{\left(1-\frac{1}{x^2}\right)dx}{\left(x+\frac{1}{x}\right)^2+3}=\frac{1}{2}\int\frac{dt}{t^2+\left(\sqrt{7}\right)^2}-\frac{1}{2}\int\frac{du}{u^2+\left(\sqrt{3}\right)^2}$

where $t=x-\frac{1}{x}$ and $u=x+\frac{1}{x}$

$I=\frac{1}{2}.\frac{1}{\sqrt{7}}\left(\tan^{-1}\frac{t}{\sqrt{7}}\right)-\frac{1}{2}.\frac{1}{\sqrt{3}}\left(\tan^{-1}\frac{u}{\sqrt{3}}\right)+C$

$=\frac{1}{2}\left[\frac{1}{\sqrt{7}}\tan^{-1}\left(\frac{x-\frac{1}{x}}{\sqrt{7}}\right)-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x+\frac{1}{x}}{\sqrt{3}}\right)\right]+C$ **Ans.**

Illustration 63

Evaluate : $\int\frac{4}{\sin^4x+\cos^4x}dx$

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Solution:

$$I = 4\int\frac{1}{\sin^4 x+\cos^4 x}dx = 4\int\frac{\sec^4 x}{1+\tan^4 x}dx = 4\int\frac{(\tan^2 x+1)\sec^2 x}{(\tan^4 x+1)}dx$$

Now, put $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$$\Rightarrow I = 4\int\frac{1+t^2}{1+t^4}dt = 4\int\frac{\frac{1}{t^2}+1}{t^2+\frac{1}{t^2}}dt$$

Now, put $t - \frac{1}{t} = z \Rightarrow \left(1+\frac{1}{t^2}\right)dt = dz$

$$\Rightarrow I = 4\int\frac{dz}{z^2+2} = \frac{4}{\sqrt{2}}\tan^{-1}\frac{z}{\sqrt{2}}+C = 2\sqrt{2}\tan^{-1}\frac{t-\frac{1}{t}}{\sqrt{2}}+C$$

$$= 2\sqrt{2}\tan^{-1}\left(\frac{\tan x-\cot x}{\sqrt{2}}\right)+C$$ **Ans.**

13. INTEGRALS OF FORM:

(i) $\int\frac{dx}{(ax+b)\sqrt{px+q}}$ and $\int\frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$; put $px + q = t^2$

(ii) $\int\frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$, put $ax + b = \frac{1}{t}$; $\int\frac{dx}{(ax^2+b)\sqrt{px^2+q}}$, put $x = \frac{1}{t}$

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Illustration 64

Evaluate $\int\frac{x+2}{(x^2+3x+3)\sqrt{x+1}}.dx$

Solution:

Let, $I = \int\frac{(x+2)dx}{(x^2+3x+3)\sqrt{x+1}}$ Put $x + 1 = t^2 \Rightarrow dx = 2tdt$

$$\therefore I = \int\frac{(t^2-1)+2}{\{(t^2-1)^2+3(t^2-1)+3\}\sqrt{t^2}}.(2t)dt = 2\int\frac{t^2+1}{t^4+t^2+1}dt = 2\int\frac{1+1/t^2}{t^2+1+1/t^2}dt$$

$$= 2\int\frac{1+1/t^2}{(t-1/t)^2+(\sqrt{3})^2}.dt = 2\int\frac{du}{u^2+(\sqrt{3})^2} \qquad \left\{\text{where } u = t-\frac{1}{t}\right\}$$

$$= \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{u}{\sqrt{3}}\right)+C = \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{t^2-1}{\sqrt{3}t}\right)+C = \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{x}{\sqrt{3(x+1)}}\right)+C$$ **Ans.**

Illustration 65

Evaluate $\int\frac{dx}{(x-1)\sqrt{x^2+x+1}}$

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Solution:

Let, $I = \int \frac{dx}{(x-1)\sqrt{x^2+x+1}}$ Put $x - 1 = \frac{1}{t} \Rightarrow dx = -1/t^2\, dt$

$$I = \int \frac{-1/t^2 dt}{1/t\sqrt{\left(\frac{1}{t}+1\right)^2+\left(\frac{1}{t}+1\right)+1}} = -\int \frac{dt}{\sqrt{3t^2+3t+1}}$$

$$= -\frac{1}{\sqrt{3}}\int \frac{dt}{\sqrt{\left(t+\frac{1}{2}\right)^2+1/12}} = -\frac{1}{\sqrt{3}}\log\left|(t+1/2)+\sqrt{(t+1/2)^2+1/12}\right|+C$$

$$= -\frac{1}{\sqrt{3}}\log\left|\left(\frac{1}{x-1}+\frac{1}{2}\right)+\sqrt{\frac{12\left(\frac{1}{x-1}+\frac{1}{2}\right)^2+1}{12}}\right|+C$$

Illustration 66

Evaluate : $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Solution:

Let, $I = \int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$. Put $x = \frac{1}{t}$ So that $dx = \frac{-1}{t^2}dt$

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$$\therefore\ I = \int \frac{-1/t^2 dt}{(1+1/t^2)\sqrt{1-1/t^2}} = -\int \frac{tdt}{(t^2+1)\sqrt{t^2-1}}$$

Again let, $t^2 = u \Rightarrow 2t\, dt = du$.

$$\therefore\ I = \frac{-1}{2}\int \frac{du}{(u+1)\sqrt{u-1}} \Rightarrow \text{Put } u - 1 = z^2 \Rightarrow du = 2z\, dz$$

$$\therefore\ I = -\frac{1}{2}\int \frac{2zdz}{(z^2+1+1)\sqrt{z^2}} = -\int \frac{dz}{(z^2+2)}$$

$$I = -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{z}{\sqrt{2}}\right)+C$$

$$I = -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{u-1}}{\sqrt{2}}\right)+C = -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{t^2-1}}{\sqrt{2}}\right)+C$$

$$= -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{2}x}\right)+C$$

Illustration 67

Evaluate $\int \frac{dx}{(x-3)^3\sqrt{x^2-6x+10}}$

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Solution:

Substituting $(x-3) = 1/t \Rightarrow dx = -1/t^2 dt$

we get, $\int \frac{dx}{(x-3)^3\sqrt{x^2-6x+10}}$

$$= -\int \frac{t^2 dt}{\sqrt{1+t^2}} = \int \frac{dt}{\sqrt{1+t^2}} - \int \sqrt{1+t^2}\,dt$$

$$= \ln\left(t+\sqrt{1+t^2}\right) - \frac{t}{2}\sqrt{1+t^2} - \frac{1}{2}\ln\left(t+\sqrt{1+t^2}\right) + C$$

$$= \frac{1}{2}\ln\left(t+\sqrt{1+t^2}\right) - \frac{t}{2}\sqrt{1+t^2} + C$$

$$= \frac{1}{2}\left(\ln\left(\frac{1+\sqrt{x^2-6x+10}}{|x-3|}\right)\right) - \frac{\sqrt{x^2-6x+10}}{2(x-3)^2} + C$$

Illustration 68

Evaluate $\int \frac{dx}{(x^2-9)\sqrt{x-1}}$

Solution:

Put $x - 1 = t^2 \Rightarrow dx = 2t\,dt$

$$\therefore I = \int \frac{dx}{(x^2-9)\sqrt{x-1}} = \int \frac{2t\,dt}{(t^2-2)(t^2+4)(t)}$$

$$= \frac{1}{3}\int \frac{t^2+4-(t^2-2)}{(t^2+4)(t^2-2)}dt = \frac{1}{3}\int\left(\frac{1}{t^2-2} - \frac{1}{t^2+4}\right)dt$$

$$= \frac{1}{6\sqrt{2}}\ell n\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right| - \frac{1}{6}\tan^{-1}\frac{t}{2} + C$$

$$= \frac{1}{6\sqrt{2}}\ell n\left|\frac{\sqrt{x-1}-\sqrt{2}}{\sqrt{x-1}+\sqrt{2}}\right| - \frac{1}{6}\tan^{-1}\left(\frac{\sqrt{x-1}}{2}\right) + C$$

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14. MANIPULATING INTEGRANDS:

(i) $\int \frac{dx}{x(x^n+1)}$, $n \in \mathbb{N}$, take x^n common & put $1 + x^{-n} = t$.

(ii) $\int \frac{dx}{x^2(x^n+1)^{(n-1)/n}}$, $n \in \mathbb{N}$, take x^n common & put $1 + x^{-n} = t^n$

(iii) $\int \frac{dx}{x^n(1+x^n)^{1/n}}$, $n \in \mathbb{N}$, take x^n common and put $1 + x^{-n} = t^n$.

Illustration 69

Evaluate : $\int \frac{dx}{x^n(1+x^n)^{1/n}}$

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Solution:

Let $I = \int \frac{dx}{x^n(1+x^n)^{1/n}} = \int \frac{dx}{x^{n+1}\left(1+\frac{1}{x^n}\right)^{1/n}}$

Put $1+\frac{1}{x^n} = t^n$, then $\frac{1}{x^{n+1}}dx = -t^{n-1}dt$

$I = -\int \frac{t^{n-1}dt}{t} = -\int t^{n-2}dt = -\frac{t^{n-1}}{n-1} + C = \frac{-1}{n-1}\left(1+\frac{1}{x^n}\right)^{\frac{n-1}{n}} + C$ **Ans.**

15. TRIGONOMETRIC INTEGRALS OF FORM:

$\int \frac{dx}{a+b\sin^2 x}$ OR $\int \frac{dx}{a+b\cos^2 x}$ OR $\int \frac{dx}{a\sin^2 x + b\sin x\cos x + c\cos^2 x}$

Divide N^r and D^r by $\cos^2 x$ and put $\tan x = t$

Illustration 70

Evaluate : $\int \frac{dx}{2+\sin^2 x}$

Solution:

Divide numerator and denominator by $\cos^2 x$

$I = \int \frac{\sec^2 x dx}{2\sec^2 x + \tan^2 x} = \int \frac{\sec^2 x dx}{2+3\tan^2 x}$

Let $\sqrt{3}\tan x = t \Rightarrow \sqrt{3}\sec^2 x\,dx = dt$

So $I = \frac{1}{\sqrt{3}}\int \frac{dt}{2+t^2} = \frac{1}{\sqrt{3}}.\frac{1}{\sqrt{2}}\tan^{-1}\frac{t}{\sqrt{2}} + C = \frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{\sqrt{3}\tan x}{\sqrt{2}}\right) + C$ **Ans.**

Illustration 71

Evaluate : $\int \frac{dx}{(2\sin x + 3\cos x)^2}$

Solution:

$\int \frac{dx}{4\sin^2 x + 12\sin x\cos x + 9\cos^2 x}$

Divide numerator and denominator by $\cos^2 x$

$\therefore\ I = \int \frac{\sec^2 x dx}{4\tan^2 x + 12\tan x + 9} = \int \frac{\sec^2 x\,dx}{(2\tan x+3)^2}$

Let $2\tan x + 3 = t \Rightarrow 2\sec^2 x dx = dt$

$I = \frac{1}{2}\int \frac{dt}{t^2} = -\frac{1}{2t} + C = -\frac{1}{2(2\tan x+3)} + C$ **Ans.**

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16. TRIGONOMETRIC INTEGRALS OF FORM:

$\int \frac{dx}{a+b\sin x}$ **OR** $\int \frac{dx}{a+b\cos x}$ **OR** $\int \frac{dx}{a+b\sin x+c\cos x}$

Convert sines and cosines into their respective tangents of the half the angles and put $\tan\frac{x}{2}=t$

Remember the following results, when putting $\tan\frac{x}{2}=t$

$$\sin x=\frac{2t}{1+t^2},\cos x=\frac{1-t^2}{1+t^2},x=2\tan^{-1}t;dx=\frac{2dt}{1+t^2}$$

Illustration 72

Evaluate $\int \frac{dx}{3+4\sin 2x}$

Solution:

Let $I=\int \frac{dx}{3+4\sin 2x}=\int \frac{dx}{3+\frac{4.2\tan x}{1+\tan^2 x}}=\int \frac{(1+\tan^2 x)dx}{3+3\tan^2 x+8\tan x}$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

$$\therefore\ I=\int \frac{dt}{3t^2+8t+3}=\frac{1}{3}\int \frac{dt}{\left(t+\frac{4}{3}\right)^2-\frac{7}{9}}$$

$$=\frac{1}{3}.\frac{3}{2\sqrt{7}}\ell n\left|\frac{t+\frac{4}{3}-\frac{\sqrt{7}}{3}}{t+\frac{4}{3}+\frac{\sqrt{7}}{3}}\right|+C=\frac{1}{2\sqrt{7}}\ell n\left|\frac{\tan x+\frac{4}{3}-\frac{\sqrt{7}}{3}}{\tan x+\frac{4}{3}+\frac{\sqrt{7}}{3}}\right|+C$$

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Illustration 73

Evaluate :

$\int \frac{dx}{3\sin x+4\cos x}$

Solution:

Let $I=\int \frac{dx}{3\sin x+4\cos x}=\int \frac{dx}{3\left\{\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right\}+4\left\{\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right\}}=\int \frac{\sec^2\frac{x}{2}dx}{4+6\tan\frac{x}{2}-4\tan^2\frac{x}{2}}$

Put $\tan\frac{x}{2}=t\Rightarrow\frac{1}{2}\sec^2\frac{x}{2}dx=dt$

So $I=\int \frac{2dt}{4+6t-4t^2}=\frac{1}{2}\int \frac{dt}{1-\left(t^2-\frac{3}{2}t\right)}=\frac{1}{2}\int \frac{dt}{\frac{25}{16}-\left(t-\frac{3}{4}\right)^2}$

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$$= \frac{1}{2}\cdot\frac{1}{2\left(\frac{5}{4}\right)}\ell n\left|\frac{\frac{5}{4}+\left(t-\frac{3}{4}\right)}{\frac{5}{4}-\left(t-\frac{3}{4}\right)}\right|+C = \frac{1}{5}\ell n\left|\frac{1+2\tan\frac{x}{2}}{4-2\tan\frac{x}{2}}\right|+C \quad \textbf{Ans.}$$

17. TRIGONOMETRIC INTEGRALS OF FORM:

$$\int\frac{a\cos x+b\sin x+c}{p\cos x+q\sin x+r}dx$$

Express Numerator $(N^r) = \ell(D^r) + m\frac{d}{dx}(D^r) + n$ and proceed.

Illustration 74

Evaluate : $\int\frac{2+3\cos\theta}{\sin\theta+2\cos\theta+3}d\theta$

Solution:

Write the Numerator = ℓ(denominator) + m(*d.c.* of denominator) + n

$\Rightarrow 2 + 3\cos\theta = \ell(\sin\theta + 2\cos\theta + 3) + m(\cos\theta - 2\sin\theta) + n.$

Comparing the coefficients of sinθ, cosθ and constant terms, we get

$3\ell + n = 2, 2\ell + m = 3, \ell - 2m = 0 \Rightarrow \ell = 6/5, m = 3/5$ and $n = -8/5$

Hence $I = \int\frac{6}{5}d\theta+\frac{3}{5}\int\frac{\cos\theta-2\sin\theta}{\sin\theta+2\cos\theta+3}d\theta-\frac{8}{5}\int\frac{d\theta}{\sin\theta+2\cos\theta+3}$

$= \frac{6}{5}\theta+\frac{3}{5}\ell n|\sin\theta+2\cos\theta+3| - \frac{8}{5}I_3$ where $I_3 = \int\frac{d\theta}{\sin\theta+2\cos\theta+3}$

In I_3, put $\tan\frac{\theta}{2}=t \Rightarrow \sec^2\frac{\theta}{2}d\theta = 2dt$

$$I_3 = 2\int\frac{dt}{t^2+2t+5} = 2\int\frac{dt}{(t+1)^2+2^2} = 2.\tan^{-1}\left(\frac{t+1}{2}\right) = \tan^{-1}\left(\frac{\tan\theta/2+1}{2}\right)$$

Hence $I = \frac{6\theta}{5}+\frac{3}{5}\ell n|\sin\theta+2\cos\theta+3|-\frac{8}{5}\tan^{-1}\left(\frac{\tan\theta/2+1}{2}\right)+C$

18. TRIGONOMETRIC INTEGRALS OF FORM:

$$\int\frac{\sin x+\cos x}{f(\sin 2x)}dx, \int\frac{\sin x-\cos x}{f(\sin 2x)}dx, \int\frac{\sin x}{f(\sin 2x)}dx, \int\frac{\cos x}{f(\sin 2x)}dx$$

Concept involved here is based on the following manipulation :

$\sin 2x = (\sin x + \cos x)^2 - 1$; $\sin 2x = 1 - (\sin x - \cos x)^2$.

The ability to convert $\sin 2x$ into a function of $\sin x + \cos x$ or $\sin x - \cos x$ paves the way for substitution in case integrand contains derivative of any of them i.e. $\cos x - \sin x$ or $\cos x + \sin x$ respectively. Also, note that

$$\sin x = \frac{\sin x+\cos x}{2}+\frac{\sin x-\cos x}{2}$$

$$\cos x = \frac{\sin x+\cos x}{2}-\frac{\sin x-\cos x}{2}$$

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Illustration 75

Find $\int \frac{\sin x+\cos x}{9+16\sin 2x}dx$

Solution:

Let $I=\int \frac{\sin x+\cos x}{9+16\sin 2x}dx$

Put $\sin x-\cos x=t \Rightarrow \sin 2x=1-t^2$ and $(\sin x+\cos x)\,dx=dt$

$\therefore\ I=\int \frac{dt}{25-16t^2}=\frac{1}{2\times 5\times 4}\ell n\left|\frac{5+4t}{5-4t}\right|+C\ =\frac{1}{40}\ell n\left|\frac{5+4\sin x-4\cos x}{5-4\sin x+4\cos x}\right|+C$

Illustration 76

Find $\int \frac{\sin x}{1+\sin x\cos x}dx$

Solution:

$$I=\int \frac{\sin x}{1+\sin x\cos x}dx=\int \frac{\sin x+\cos x+\sin x-\cos x}{2+\sin 2x}dx$$

$$=\int \frac{\sin x+\cos x}{3-(\sin x-\cos x)^2}dx-\int \frac{\cos x-\sin x}{1+(\cos x+\sin x)^2}dx$$

$$=\int \frac{dt}{3-t^2}-\int \frac{du}{1+u^2}\quad \begin{bmatrix} t=\sin x-\cos x \\ u=\sin x+\cos x \end{bmatrix}\ =\frac{1}{2\sqrt{3}}\ell n\left|\frac{t+\sqrt{3}}{t-\sqrt{3}}\right|-\tan^{-1}u+C$$

$$=\frac{1}{2\sqrt{3}}\ell n\left|\frac{\sin x-\cos x+\sqrt{3}}{\sin x-\cos x-\sqrt{3}}\right|-\tan^{-1}(\sin x+\cos x)+C$$

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19. USE OF PARTIAL FRACTIONS:

Let $p(x)$ and $q(x)$ be polynomial functions and D_f denote degree of any polynomial $f(x)$. A function of the form

$$\frac{p(x)}{q(x)};q(x)\neq 0$$

is called a Rational function. If $D_p < D_q$, then rational function is said to be proper, otherwise it is called improper In case of improper rational function by using long division method, we get two polynomials $Q(x)$ and $R(x)$ such that

$$\frac{p(x)}{q(x)}=Q(x)+\frac{R(x)}{q(x)};D_R<D_q$$

If polynomial $q(x) = q_1(x).\ q_2(x)$, where q_1 and q_2 are coprime polynomials, then there exists polynomial R_1 and R_2 such that

$$\frac{R(x)}{q(x)}=\frac{R_1(x)}{q_1(x)}+\frac{R_2(x)}{q_2(x)}\ \text{and}\ D_{R_i}<D_{q_i}\ \text{for}\ i=1,2$$

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Form of the rational function	Form of the partial fraction
$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$
$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$
$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$

Illustration 77

Evaluate : $\int\frac{x}{(x-2)(x+5)}dx$

Solution:

Let $\frac{x}{(x-2)(x+5)}=\frac{A}{x-2}+\frac{B}{x+5}$

or $x = A(x + 5) + B(x - 2)$.

by comparing the coefficients, we get

$A = 2/7$ and $B = 5/7$ so that

$$\int\frac{x}{(x-2)(x+5)}dx=\frac{2}{7}\int\frac{dx}{x-2}+\frac{5}{7}\int\frac{dx}{x+5}=\frac{2}{7}\ell n|(x-2)|+\frac{5}{7}\ell n|(x+5)|+C$$ **Ans.**

Illustration 78

Evaluate $\int\frac{x^4}{(x+2)(x^2+1)}dx$

Solution:

$$\frac{x^4}{(x+2)(x^2+1)}=(x-2)+\frac{3x^2+4}{(x+2)(x^2+1)}$$

Now, $\frac{3x^2+4}{(x+2)(x^2+1)}=\frac{16}{5(x+2)}+\frac{-\frac{1}{5}x+\frac{2}{5}}{x^2+1}$

So, $\frac{x^4}{(x+2)(x^2+1)}=x-2+\frac{16}{5(x+2)}+\frac{-\frac{1}{5}x+\frac{2}{5}}{x^2+1}$

Now, $\int\frac{x^4}{(x+2)(x^2+1)}dx$

$$=\int\left((x-2)+\frac{16}{5(x+2)}+\frac{-\frac{1}{5}x+\frac{2}{5}}{x^2+1}\right)dx$$

$$=\frac{x^2}{2}-2x+\frac{2}{5}\tan^{-1}x+\frac{16}{5}\ell n|x+2|-\frac{1}{10}\ell n(x^2+1)+C$$ **Ans.**

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20. INTEGRATION BY PARTS:

Let $y = u(x)$ and $y = v(x)$ are differentiable functions of x. Then

$$\int u.v\,dx = u\int v\,dx - \int\left[\frac{du}{dx}.\int v\,dx\right]dx$$

Proof : $\frac{d}{dx}\left[f(x).g(x)\right] = f(x).g'(x) + g(x).f'(x)$

$\therefore \quad \int \underset{I}{f(x)}.\underset{II}{g'(x)}dx = f(x).g(x) - \int g(x).f'(x)dx$

Points to be noted :

- $y = u(x)$ and $y = v(x)$ are commonly called as first and second function respectively
- While using integration by parts, chose u and v such that

  $\int v\,dx$ and $\int\left(\left(\frac{du}{dx}\right).\int v\,dx\right)dx$ are easy to integrate
- **Choosing second function :** Generally the function which is easier to integrate is taken as second function. For this we usually follow ILATE rule Here I-Inverse function, L-Logarithmic function, A-Algebraic function, T-Trigonometric function & E-Exponential function.

  In order stated above, the function on the RIGHT is generally taken as second function

Integral	First function	Second function
$\int x^3 \sin x\,dx$	x^3	$\sin x$
$\int e^x \cos x\,dx$	$\cos x$	e^x
$\int (\ell nx)\sqrt{x}\,dx$	$\sqrt{x}$	ℓnx
$\int \frac{x\tan^{-1}x}{\sqrt{1+x^2}}dx$	$\tan^{-1}x$	$\frac{x}{\sqrt{1+x^2}}$

- In case there is only one function, sometimes unity (1) is taken as second function. For example,

  $\int(\ell nx)dx = \int(\ell nx)(1)dx, \int\left(\tan^{-1}x\right)dx = \int\left(\tan^{-1}x\right)(1)dx$ ...etc.

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Illustration 79

Evaluate $\int e^{\ell nx+x}.dx$

Solution:

$$\int e^{\ell nx+x}.dx = x\int e^x dx - \int(1)\left(\int e^x dx\right)dx = xe^x - \int e^x dx = xe^x - e^x + C$$

Illustration 80

Evaluate $\int x^4\,\ell nx\,dx$

Solution:

$$\int \underset{II}{x^4}\ \underset{I}{\ell nx}\,dx = \ell nx\int x^4 dx - \int\left(\frac{d(\ell nx)}{dx}\int x^4 dx\right)dx$$

$$= \ell nx\cdot\frac{x^5}{5} - \int\frac{1}{x}\times\frac{x^5}{5}dx = \frac{x^5}{5}\ell nx - \frac{x^5}{25} + C$$

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Illustration 81

Evaluate $\int x\cos 2x\,dx$

Solution:

$$\int x\cos 2x\,dx = x\int\cos 2x\,dx - \int(1)\left(\int\cos 2x\,dx\right)dx$$

$$= \frac{x\sin 2x}{2} - \int\frac{\sin 2x}{2}dx$$

$$= \frac{x\sin 2x}{2} + \frac{\cos 2x}{4} + C$$

Illustration 82

Evaluate $\int \ell n(2x+3)^{(2x+3)}\,dx$

Solution:

$$\int \ell n(2x+3)^{(2x+3)}\,dx = \int \underbrace{(2x+3)}_{II}\,\underbrace{\ln(2x+3)}_{I}\,dx$$

$$= \ell n(2x+3)\int(2x+3)\,dx - \int\frac{1}{(2x+3)}\left(\int(2x+3)\,dx\right)dx$$

$$= \ell n(2x+3)\frac{(2x+3)^2}{4} - \int\frac{1}{2x+3}\times\frac{(2x+3)^2}{2}dx$$

$$= \ell n(2x+3)\frac{(2x+3)^2}{4} - \frac{(2x+3)^2}{4} + C$$

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Illustration 83

Evaluate $\int x^2 e^x\,dx$

Solution:

$$\int x^2 e^x\,dx = x^2(e^x) - \int 2xe^x dx$$

$$x^2(e^x) - \left[2xe^x - \int 2e^x dx\right] = x^2e^x - 2xe^x + 2e^x + C$$

Illustration 84

Evaluate $\int \sin^{-1}x\,dx$

Solution: **M-1 :**

Let 1 is algebraic function

$$\int \underset{II}{1}\cdot\underset{I}{\sin^{-1}x}\,dx = \sin^{-1}x(x) - \int\left(\frac{1}{\sqrt{1-x^2}}\right)(x)\,dx$$

Put $1 - x^2 = t \Rightarrow \quad -2xdx = dt$

$$\therefore \int\sin^{-1}x\,dx = x\sin^{-1}(x) + \frac{1}{2}\int\frac{dt}{\sqrt{t}}$$

$$= x\sin^{-1}x + \frac{1}{2}\frac{\sqrt{t}}{\left(\frac{1}{2}\right)} + C = x\sin^{-1}x + \sqrt{1-x^2} + C$$

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M-2 :

Put $\sin^{-1}x = t \Rightarrow x = \sin t \Rightarrow \quad dx = \cos t\, dt$

$\therefore \int \sin^{-1}x\, dx = \int \underset{I}{t}\ \underset{II}{\cos t}\ dt = t(\sin t) - \int (1)\sin t\, dt$

$= t\sin t + \cos t + C = x\sin^{-1}x + \sqrt{1-x^2} + C$

Illustration 85

Evaluate $\int \ell nx\, dx$

Solution:

M-1 :

$\int \underset{II}{1} \cdot \underset{I}{\ell nx}\ dx = \ln(x)\cdot(x) - \int \frac{1}{x}\cdot x\, dx = x\ell nx - x + C$

M-2 :

$\int \ell nx\, dx$

Put, $\ell nx = t \Rightarrow x = e^t \Rightarrow dx = e^t dt$

$\therefore \int \ell nx\, dx = \int \underset{I}{t} \cdot \underset{II}{e^t}\ dt = t.e^t - e^t + C = x\ell nx - x + C$

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Illustration 86

Evaluate $\int \cos\sqrt{x}\, dx$

Solution:

Consider $I = \int \cos\sqrt{x}dx$

Let $\sqrt{x} = t$ then $\frac{1}{2\sqrt{x}}dx = dt$

i.e. $dx = 2\sqrt{x}dt$ or $dx = 2t\, dt.$

So $I = \int \cos t.2tdt$

Taking t as first function, integrating it by parts, we get

$I = 2\left[t\int \cos t dt - \int \left\{\frac{dt}{dt}\int \cos t dt\right\}dt\right]$

$\Rightarrow \quad I = 2\left[t\sin t - \int 1.\sin t dt\right] = 2[t\sin t + \cos t] + C$

$\Rightarrow \quad I = 2\left[\sqrt{x}\sin\sqrt{x} + \cos\sqrt{x}\right] + C$ **Ans.**

Illustration 87

Evaluate : $\int \frac{x}{1+\sin x}dx$

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Solution:

Let $I=\int\frac{x}{1+\sin x}dx=\int x\left(\frac{1-\sin x}{\cos^2 x}\right)dx$

$=\int_{I}x\underbrace{\left(\sec^2 x-\sec x\tan x\right)}_{II}dx$

$=x\left(\tan x-\sec x\right)-\int(1)\left(\tan x-\sec x\right)dx$

$=x\left(\tan x-\sec x\right)+\int\frac{1-\sin x}{\cos x}dx$

$=x\left(\tan x-\sec x\right)+\int\frac{\cos x\,dx}{1+\sin x}$

$=x\left(\tan x-\sec x\right)-\ell n|1+\sin x|+C$

Illustration 88

Prove that $\int\sqrt{x^2-a^2}dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+C$

Solution:

Let $I=\int\sqrt{x^2-a^2}dx$

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Taking constant function 1 as the second function and integration by parts, we have

$I=x\sqrt{x^2-a^2}-\int x.\frac{1}{2}\frac{2x}{\sqrt{x^2-a^2}}dx$

$=x\sqrt{x^2-a^2}-\int\frac{x^2}{\sqrt{x^2-a^2}}dx=x\sqrt{x^2-a^2}-\int\frac{x^2-a^2+a^2}{\sqrt{x^2-a^2}}dx$

$=x\sqrt{x^2-a^2}-\int\sqrt{x^2-a^2}dx-a^2\int\frac{dx}{\sqrt{x^2-a^2}}$

$=x\sqrt{x^2-a^2}-I-a^2\int\frac{dx}{\sqrt{x^2-a^2}}$

or $2I=x\sqrt{x^2-a^2}-a^2\int\frac{dx}{\sqrt{x^2-a^2}}$

or $I=\int\sqrt{x^2-a^2}dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+C$

Illustration 89

Find $\int\sqrt{a^2-x^2}dx$

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Solution:

Let $I = \int (1).\sqrt{a^2 - x^2}\, dx = \sqrt{a^2 - x^2}.x + \int \frac{x}{\sqrt{a^2 - x^2}} x dx$

$= x\sqrt{a^2 - x^2} + \int \frac{a^2 - \left(a^2 - x^2\right)}{\sqrt{a^2 - x^2}} dx$

$\Rightarrow I = x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\frac{x}{a} - I$

$\Rightarrow I = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

21. TWO CLASSIC INEGRALS:

(a) $\int e^x (f(x) + f'(x))dx = e^x f(x) + C$

Proof:

Let $I = \int e^x [f(x) + f'(x)]dx$

$\Rightarrow I = \int e^x f(x)dx + \int e^x f'(x)dx$

Integrate the first integral on the RHS by parts taking e^x as the second function, we get

$I = e^x f(x) - \int e^x f'(x)dx + \int e^x f'(x)dx$

$\Rightarrow I = e^x f(x) + C$

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Thus to evaluate the integrals of the type $\int e^x g(x)dx$, we first try to express $g(x)$ as the sum of the function and its derivative i.e., $g(x) = f(x) + f'(x)$ and then we use the result derived above.

(b) $\int (f(x) + xf'(x))dx = x f(x) + C$

Proof : $\int (f(x) + xf'(x))dx = x f(x) + c = \int f(x)\cdot 1 dx + \int xf'(x)dx$

$= \left[f(x)\cdot x - \int f'(x)\cdot x dx \right] + \int xf'(x)dx$

$= x f(x) + C$

Illustration 90

Evaluate $\int e^x (\sin x + \cos x)dx$

Solution:

$\int e^x (\sin x + \cos x)dx = e^x \sin x + C$

$f(x)$ $f'(x)$

Illustration 91

Evaluate $\int e^x \left(\frac{2x^2 + x^3}{x}\right)dx$

Solution:

$\int e^x \left(\underset{f'(x)}{2x} + \underset{f(x)}{x^2}\right)dx = e^x.x^2 + C$

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Illustration 92

Evaluate $\int \frac{xe^x}{(1+x)^2}dx$

Solution:

$$\int e^x\left(\frac{x+1}{(x+1)^2}+\frac{(-1)}{(x+1)^2}\right)dx$$

$$= \int e^x\left(\underbrace{\frac{1}{x+1}}_{f(x)}+\underbrace{\frac{(-1)}{(x+1)^2}}_{f'(x)}\right)dx$$

$$= e^x\cdot\frac{1}{1+x}+C$$

Illustration 93

Evaluate $\int e^x\left(\frac{\sin 2x-2}{1-\cos 2x}\right)dx$

Solution:

$$\int e^x\left(\frac{2\sin x\cos x-2}{2\sin^2 x}\right)= \int e^x\left(\frac{2\sin x\cos x}{2\sin^2 x}-\frac{2}{2\sin^2 x}\right)dx$$

$$\int e^x(\underbrace{\cot x}_{f(x)}+\underbrace{(-\operatorname{cosec}^2 x)}_{f'(x)})dx = e^x\cot x + C$$

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Illustration 94

Evaluate $\int e^x\left(\frac{1-x}{1+x^2}\right)^2 dx$

Solution:

$$\int e^x\left(\frac{1-x}{1+x^2}\right)^2 dx = \int e^x\frac{(1-2x+x^2)}{(1+x^2)^2}dx$$

$$= \int e^x\left(\frac{1}{(1+x^2)}-\frac{2x}{(1+x^2)^2}\right)dx = \frac{e^x}{1+x^2}+C \qquad \textbf{Ans.}$$

Illustration 95

Evaluate $\int e^x\left(\frac{x^4+2}{(1+x^2)^{5/2}}\right)dx$

Solution:

Let $I = \int e^x\left(\frac{x^4+2}{(1+x^2)^{5/2}}\right)dx = \int e^x\left(\frac{1}{(1+x^2)^{1/2}}+\frac{1-2x^2}{(1+x^2)^{5/2}}\right).dx$

$$= \int e^x\left(\frac{1}{(1+x^2)^{1/2}}-\frac{x}{(1+x^2)^{3/2}}+\frac{x}{(1+x^2)^{3/2}}+\frac{1-2x^2}{(1+x^2)^{5/2}}\right)dx$$

$$= \frac{e^x}{(1+x^2)^{1/2}}+\frac{xe^x}{(1+x^2)^{3/2}}+C = \frac{e^x\{1+x^2+x\}}{(1+x^2)^{3/2}}+C$$

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Illustration 96

Evaluate $\int(\sin x + x\cos x)dx$

Solution:

$$\int(\sin x + x\cos x)dx = \int \underset{f(x)}{(\sin x + x\cos x)}\, dx = x\ \sin x + C$$

Illustration 97

Evaluate $\int\left(\ell n(\ell nx^2) + \frac{2}{\ln x^2}\right)dx$

Solution:

$$\int\left(\ell n(\ell nx^2) + \frac{2}{\ell nx^2}\right)dx = \int(\underbrace{\ell n\,(2\ell nx)}_{f(x)} + \underbrace{x\left(\frac{1}{2\ell nx}.\frac{2}{x}\right)}_{f'(x)})dx = x\ \ell n\ (\ell nx^2) + C$$

Illustration 98

Evaluate $\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx$

Solution:

$$\int e^{\tan^{-1}x}\left(\frac{1+x^2}{1+x^2} + \frac{x}{1+x^2}\right)dx = \int e^{\tan^{-1}x}\left(1 + \frac{x}{1+x^2}\right)dx$$

$$= \int \underset{f(x)}{e^{\tan^{-1}x}} + x\cdot\underbrace{\frac{e^{\tan^{-1}x}}{1+x^2}}_{x\,f'(x)}\, dx = xe^{\tan^{-1}x} + C$$

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Illustration 99

Evaluate $\int\frac{x+\sin x}{1+\cos x}dx$

Solution:

$$I = \int\frac{x+\sin x}{1+\cos x}dx = \int\left(\frac{x+\sin x}{2\cos^2\frac{x}{2}}\right)dx = \int\left(x\frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}\right)dx = x\ \tan\ \frac{x}{2} + C$$ **Ans.**

Illustration 100 :

Prove that $\int e^{g(x)}\left(g'(x).f(x) + f'(x)\right)dx = e^{g(x)}.f(x)$.

Solution:

$$I = \int e^{g(x)}(g'(x)f(x) + f'(x))dx$$

$$= \int e^{g(x)}g'(x).f(x)dx + \int e^{g(x)}f'(x)dx$$

Integrate the first integral on the R.H.S. by parts taking $e^{g(x)}.g'(x)$ as the second function, we get

$$I = e^{g(x)}f(x) - \int f'(x)e^{g(x)}dx + \int e^{g(x)}.f'(x)dx$$

$$= e^{g(x)}.f(x)$$

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Illustration 101

Evaluate $\int e^{\tan x}(\sec x - \sin x)dx$

Solution:

$$\int e^{\tan x}(\sec x - \sin x)dx$$

$$= \int \underset{e^g}{e^{\tan x}} \left(\underset{g'}{\sec^2 x} . \underset{f}{\cos x} + \underbrace{(-\sin x)}_{f'} \right) dx$$

$= e^{\tan x}.\cos x + C$

Illustration 102

$$\int e^{ax}.\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$$

and

$$\int e^{ax}.\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$$

Solution:

Let $I = \int e^{ax}\sin bx\,dx$

Integrate by parts taking e^{ax} as the first function, we get

$$I = \frac{-e^{ax}\cos bx}{b} - \int ae^{ax}\left(\frac{-\cos bx}{b}\right)dx$$

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On integrating the second term by parts again, we get

$$I = -\frac{1}{b}e^{ax}\cos bx + \frac{a}{b}\left[\frac{1}{b}e^{ax}\sin bx - \int \frac{a}{b}e^{ax}\sin bx\,dx\right]$$

$$\Rightarrow I = -\frac{1}{b}e^{ax}\cos bx + \frac{a}{b^2}e^{ax}\sin bx - \frac{a^2}{b^2}I \Rightarrow \left(1+\frac{a^2}{b^2}\right)I = \frac{e^{ax}}{b^2}\ (a\sin bx - b\cos bx) + C$$

$$\Rightarrow I = \frac{e^{ax}}{a^2+b^2}\ (a\sin bx - b\cos bx) + C$$

Similarly we can show that $\int e^{ax}\cos bx = \frac{e^{ax}}{a^2+b^2}\ (a\cos bx + b\sin bx) + C$

Illustration 103

Evaluate $\int e^x \sin x dx$

Solution:

$$I = \int e^x.\sin x\,dx = \sin x.e^x - \int \cos x.e^x dx$$

$$= \sin x e^x - \left[\cos x e^x - \int(-\sin x)e^x dx\right] = \sin x\ e^x - \cos x\ e^x - I$$

$$\Rightarrow 2I = \sin x\ e^x - \cos x\ e^x$$

$$\Rightarrow I = \frac{1}{2}(\sin x - \cos x)e^x + c$$

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Illustration 104

$\int e^{2x}\cos 3x dx = e^{2x}\{A\cos 3x + B\sin 3x\} + C$. Find A & B

Solution:

Method-I : Use formula $\int e^{ax}.\cos bx dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$, we get

$\int e^{2x}\cos 3x dx = \frac{e^{2x}}{13}[2\cos 3x + 3\sin 3x] + C$

Method-II : Differentiate both sides

$e^{2x}\cos 3x = e^{2x}(-3A\sin 3x + 3B\cos 3x) + 2e^{2x}(A\cos 3x + B\sin 3x)$

$= e^{2x}[(2A+3B)\cos 3x + (2B-3A)\sin 3x]$

$\Rightarrow$ $2A + 3B = 1$

$2B - 3A = 0$

$\Rightarrow$ $A = \frac{2}{13}$ and $B = \frac{3}{13}$ **Ans.**

22. REDUCTION FORMULAE:

Observe the following illustrations :

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Illustration 105

If $I_n = \int (\ell nx)^n dx$ then prove that $I_n + nI_{n-1} = x(\ell nx)^n$

Solution:

$I_n = x(\ell nx)^n - \int \frac{x(n)(\ell nx)^{n-1}}{x} dx = x(\ell nx)^n - nI_{(n-1)}$

$\Rightarrow I_n + nI_{n-1} = x(\ell nx)^n$

Illustration 106

Prove that $\int \sin^n x dx = -\frac{1}{n}\sin^{n-1} x.\cos x + \frac{n-1}{n}\int \sin^{n-2} x dx$

Solution:

If we take $dv = \sin x\, dx$, then we have $v = -\cos x$ and we may integrate by parts with

$u = \sin^{n-1} x$, $du = (n-1)\sin^{n-2} x.\cos x$.

Using the fact that $\sin^2 x + \cos^2 x = 1$, one may thus conclude that

$\int \sin^n x dx = -\sin^{n-1} x.\cos x + (n-1)\int \sin^{n-2} x.\cos^2 x dx$

$= -\sin^{n-1} x.\cos x + (n-1)\int \sin^{n-2} x.(1-\sin^2 x) dx$

$= -\sin^{n-1} x.\cos x + (n-1)\int \sin^{n-2} x dx + (1-n)\int \sin^n x dx$

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Illustration 107

$$\int \sec^n x dx = \frac{1}{n-1}\sec^{n-2} x.\tan x + \frac{n-2}{n-1}\int \sec^{n-2} x dx$$

Solution:

In this case, we note that $(\tan x)' = \sec^2 x$ and we write the given integral as

$$\int \sec^n x dx = \int \sec^{n-2} x.\sec^2 x dx$$

If we take $dv = \sec^2 x dx$, then we have $v = \tan x$ and we may integrate by parts with

$u = \sec^{n-2} x$, $du = (n-2)\sec^{n-3} x.\sec x \tan x = (n-2)\sec^{n-2} x.\tan x$.

Using the fact that $1 + \tan^2 x = \sec^2 x$, one may thus establish the identity

$$\int \sec^n x dx = \sec^{n-2} x.\tan x - (n-2)\int \sec^{n-2} x.\tan^2 x dx$$

$$= \sec^{n-2} x.\tan x - (n-2)\int \sec^{n-2} x.(\sec^2 x - 1)dx$$

$$= \sec^{n-2} x.\tan x - (n-2)\int \sec^n x dx + (n-2)\int \sec^{n-2} x dx$$

Since the integral on the left hand side also appears on the right hand side, this gives

$$(n-1)\int \sec^n x dx = \sec^{n-2} x.\tan x + (n-2)\int \sec^{n-2} x dx$$

23. MISCELLANEOUS EXAMPLES:

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Illustration 108

Evaluate $\int \frac{\cos^4 x dx}{\sin^3 x\{\sin^5 x + \cos^5 x\}^{\frac{3}{5}}}$

Solution:

$$I = \int \frac{\cos^4 x}{\sin^3 x\{\sin^5 x + \cos^5 x\}^{\frac{3}{5}}} dx = \int \frac{\cos^4 x}{\sin^6 x\{1 + \cot^5 x\}^{\frac{3}{5}}} dx = \int \frac{\cot^4 x \operatorname{cosec}^2 x dx}{(1+\cot^5 x)^{3/5}}$$

Put $1 + \cot^5 x = t \Rightarrow 5\cot^4 x \operatorname{cosec}^2 x dx = -dt$

$$\therefore I = -\frac{1}{5}\int \frac{dt}{t^{3/5}} = -\frac{1}{2}t^{2/5} + C = -\frac{1}{2}(1+\cot^5 x)^{2/5} + C$$ **Ans.**

Illustration 109

Evaluate : $\int \frac{2\sin 2x - \cos x}{6 - \cos^2 x - 4\sin x} dx$

Solution:

$$I = \int \frac{2\sin 2x - \cos x}{6 - \cos^2 x - 4\sin x} dx = \int \frac{(4\sin x - 1)\cos x}{6 - (1 - \sin^2 x) - 4\sin x} dx = \int \frac{(4\sin x - 1)\cos x}{\sin^2 x - 4\sin x + 5} dx$$

Put $\sin x = t$, so that $\cos x\, dx = dt$.

$$\therefore I = \int \frac{(4t-1)dt}{(t^2 - 4t + 5)} \quad \ldots(i)$$

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Now, let $(4t - 1) = \lambda(2t - 4) + \mu$

Comparing coefficients of like powers of t, we get

$2\lambda = 4, -4\lambda + \mu = -1$ ...(ii)

$\Rightarrow \lambda = 2, \mu = 7$

$\therefore I = \int \frac{2(2t-4)+7}{t^2-4t+5}dt$ {using (i) and (ii)}

$= 2\int \frac{2t-4}{t^2-4t+5}dt + 7\int \frac{dt}{t^2-4t+5} = 2\log|t^2-4t+5| + 7\int \frac{dt}{t^2-4t+4-4+5}$

$= 2\log|t^2-4t+5| + 7\int \frac{dt}{(t-2)^2+(1)^2} = 2\log|t^2 - 4t + 5| + 7.\tan^{-1}(t - 2) + C$

$= 2\log|\sin^2 x - 4\sin x + 5| + 7\tan^{-1}(\sin x - 2) + C.$ **Ans.**

Illustration 110

Evaluate $\int \sqrt{\frac{3-x}{3+x}}.\sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-x}\right)dx$

Solution:

Here, $I = \int \sqrt{\frac{3-x}{3+x}}.\sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-x}\right)dx$

Put $x = 3\cos 2\theta \Rightarrow dx = -6\sin 2\theta d\theta$

$= \int \sqrt{\frac{3-3\cos 2\theta}{3+3\cos 2\theta}}.\sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-3\cos 2\theta}\right)(-6\sin 2\theta)d\theta$

$\therefore I = \int \frac{\sin\theta}{\cos\theta}.\sin^{-1}(\sin\theta).(-6\sin 2\theta)d\theta = -6\int \theta.(2\sin^2\theta)d\theta$

$= -6\int \theta(1-\cos 2\theta)d\theta = -6\left\{\frac{\theta^2}{2} - \int \theta\cos 2\theta\, d\theta\right\} + C$

$= -6\left\{\frac{\theta^2}{2} - \left(\theta\frac{\sin 2\theta}{2} - \int 1.\left(\frac{\sin 2\theta}{2}\right)d\theta\right)\right\} + C$

$= -3\theta^2 + 6\left\{\theta\frac{\sin 2\theta}{2} + \frac{\cos 2\theta}{4}\right\} + C$

$= \frac{1}{4}\left\{-3\left(\cos^{-1}\left(\frac{x}{3}\right)\right)^2 + 2\sqrt{9-x^2}.\cos^{-1}\left(\frac{x}{3}\right) + 2x\right\} + C$

Illustration 111

Evaluate : $\int \frac{\tan\left(\frac{\pi}{4} - x\right)}{\cos^2 x\sqrt{\tan^3 x + \tan^2 x + \tan x}}dx$

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Solution:

$$I = \int \frac{\tan\left(\frac{\pi}{4}-x\right)}{\cos^2 x\sqrt{\tan^3 x+\tan^2 x+\tan x}}dx = \int \frac{(1-\tan^2 x)dx}{(1+\tan x)^2\cos^2 x\sqrt{\tan^3 x+\tan^2 x+\tan x}}$$

$$I = \int \frac{-\left(1-\frac{1}{\tan^2 x}\right)\sec^2 x dx}{\left(\tan x+2+\frac{1}{\tan x}\right)\sqrt{\tan x+1+\frac{1}{\tan x}}}$$

Let $y = \sqrt{\tan x+1+\frac{1}{\tan x}} \Rightarrow 2y\,dy = \left(\sec^2 x-\frac{1}{\tan^2 x}.\sec^2 x\right)dx$

$\therefore \quad I = \int \frac{-2ydy}{(y^2+1).y} = -2\int \frac{dy}{1+y^2}$

$= -2\tan^{-1}y + c = -2\tan^{-1}\left(\sqrt{\tan x+1+\frac{1}{\tan x}}\right)+C$ **Ans.**

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STANDARD RESULTS

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C\,;\, n \neq -1$

2. $\int \frac{1}{x} dx = \ell n|x| + C$

3. $\int \cos x dx = \sin x + C$

4. $\int \sin x dx = -\cos x + C$

5. $\int \sec^2 x dx = \tan x + C$

6. $\int \text{cosec}^2 x\, dx = -\cot x + C$

7. $\int \sec x \tan x\, dx = \sec x + C$

8. $\int \text{cosec}x\, \cot x\, dx = -\text{cosec}x + C$

9. $\int a^x dx = \frac{a^x}{\ell na} + C \quad (a > 0, a \neq 0)$

10. $\int e^x dx = e^x + C$

11. $\int \text{cosec}\, x.\cot x.dx = -\text{cosec}\, x + c$

12. $\int \sec x.\tan x.\, dx = \sec x + c$

13. $\int \tan x\, dx = \ell n|\sec x| + c$

14. $\int \cot x\, dx = -\ell n|\text{cosec}\, x| + c$

15. $\int \sec x\, dx = \ell n|\sec x + \tan x| + c = \ell n \tan\left|\frac{\pi}{4} + \frac{x}{2}\right| + c$

16. $\int \text{cosec}\, x\, dx = -\ell n|\text{cosec}\, x + \cot x| + C = \ell n|\text{cosec}\, x - \cot x| + c = \ell n\left|\tan\frac{x}{2}\right| + c$

17. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a} + c\,,\, x > 0$

18. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ell n\left|\frac{x-a}{x+a}\right| + c$

19. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

20. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + c$

21. $\int \frac{dx}{\sqrt{x^2 + a^2}} = \ell n\left[x + \sqrt{x^2 + a^2}\right] + c$

22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \ell n\left|x + \sqrt{x^2 - a^2}\right| + c$

23. $\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$

24. $\int \sqrt{x^2 + a^2} dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ell n\left(x + \sqrt{x^2 + a^2}\right) + c$

25. $\int \sqrt{x^2 - a^2} dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ell n\left|x + \sqrt{x^2 - a^2}\right| + c$

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